

Modular Forms

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0 Introduction

What are modular forms? They are analytic objects arising from a geometric context:

$$\mathbb{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$$

the complex upper-half plane. We care about the group action acting on the plane for example:

$$GL_2^+(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc > 0 \right\}$$

Definition 0.1. Recall that the group action is defined by: Let G be a group and S a set. A (left) action of G on S is a function " $\cdot$ " : $G \times S \rightarrow S$ such that it satisfies the following conditions:

- (i) $(g_1 g_2) \cdot s = g_1 \cdot (g_2 \cdot s)$ for all $g_1, g_2 \in G$ and $s \in S$.
- (ii) $e \cdot s = s$ for all $s \in S$.

Example. In our case,

$$\cdot : GL_2^+(\mathbb{R}) \times \mathbb{H} \rightarrow \mathbb{H}$$

$$\gamma z \mapsto \gamma \cdot z = \frac{az + b}{cz + d}$$

where $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Note that this is well-defined since

$$\text{Im}(\gamma z) = \frac{\det(\gamma) \text{Im}(z)}{|cz + d|^2} > 0$$

Also $cz + d \neq 0$ for all $z \in \mathbb{H}$ since $\det(\gamma) \neq 0$. To check this group action is well-defined is an exercise for the reader.

Example. Scalar matrices act trivially. Take $\gamma = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, a \in \mathbb{R}_{\neq 0}$. Then

$$\gamma z = \frac{az + 0}{0 + a} = z$$

In addition,

$$SL_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2^+(\mathbb{R}) \mid ad - bc = 1 \right\}$$

also acts on $\mathbb{H}$, these are the transformations that preserve distances.

We will restrict ourselves to the action of certain discrete subgroups of $SL_2(\mathbb{R})$ such as

$$SL_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, ad - bc = 1 \right\}$$

known as the full modular group. And some of its subgroups of finite index. For these groups denoted by Γ , the orbit spaces the quotient space i.e.

$$Y(\Gamma) := \Gamma \backslash \mathbb{H} = \{\Gamma z \mid z \in \mathbb{H}\}$$

which means $z \sim_{\Gamma} z'$ if $\exists \gamma \in \Gamma$ such that $z' = \gamma z$. The orbit spaces are in fact:

- (1) Complex manifolds, connected (Riemann Surface).
- (2) Algebraic curve called the modular curves.
- (3) Moduli spaces, parametrize elliptic curves with special structure

To study these spaces, we would like to know nice functions them, they would have to be invariant under Γ . Instead, we introduce an extra method—this is where modular forms course in.

Definition 0.2 (A modular form for $SL_2(\mathbb{Z})$). Let $k \in \mathbb{Z}$. A modular form of weight k for $SL_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ satisfying two properties:

- (i) (Modularity condition)

$$f\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} z\right) = (cz + d)^k f(z)$$

for all $z \in \mathbb{H}$ and all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$.

- (ii) (Conditions at ∞) $f(z)$ is bounded as $\text{Im}(z) \rightarrow +\infty$.

Remark (Some remarks about $SL_2(\mathbb{Z})$).

- (1) If $\gamma \in \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ then $-\gamma \in SL_2(\mathbb{Z})$. Note that

$$\gamma z = (-\gamma)z \text{ for all } z \in \mathbb{H}$$

So we are really looking at the action of $PSL_2(\mathbb{Z}) := SL_2(\mathbb{Z})/\{\pm I\}$. In some texts, this is the modular group.

(2) Look at modularity conditions with weight k for $-I$:

$$f(z) = f\left(\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} z\right) = (0z + (-1))^k f(z) = (-1)^k f(z)$$

i.e. $f(z) = (-1)^k f(z)$ for all $z \in \mathbb{H}$. So if k is odd, $f = 0$. (there are no non-trivial modular forms for $SL_2(\mathbb{Z})$ of odd weight).

(3) $f = 0$ is a modular form of weight k , for any $k \in \mathbb{Z}$. In fact $f = c \neq 0$ constant is a modular form of weight 0.

(4) Matrices in $SL_2(\mathbb{Z})$: $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ where $Tz = z + 1$ then

$$T^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \text{ for all } n \in \mathbb{Z}$$

Also, $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ where $Sz = -\frac{1}{z}$, $z \in \mathbb{H}$. Then we have

$$S^2 = -I, \quad (ST)^3 = (TS)^3 = -I$$

A look ahead: We will prove that S and T generated $SL_2(\mathbb{Z})$. We need the Lemma below, we only need to check the modularity condition for S and T . For S ,

$$f\left(-\frac{1}{z}\right) = z^k f(z)$$

and for T , $f(z + 1) = f(z)$ for all $z \in \mathbb{H}$.

Lemma 0.3. Suppose $f : \mathbb{H} \rightarrow \mathbb{C}$, satisfies the modularity conditions of weight k for $\gamma_1, \gamma_2 \in SL_2(\mathbb{Z})$. Then f satisfies the condition for the $\gamma_1\gamma_2$ and γ_1^{-1} and hence for the group generated by γ_1 and γ_2 .

Proof. Let $\gamma_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $\gamma_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$. Note that

$$\gamma_1\gamma_2 = \begin{pmatrix} * & * \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{pmatrix}, \quad \gamma_1^{-1} = \begin{pmatrix} d_1 & -b_1 \\ -c_1 & a_1 \end{pmatrix}$$

Then

$$\begin{aligned} f((\gamma_1\gamma_2)z) &= f(\gamma_1(\gamma_2z)) \stackrel{\text{modularity condition on } \gamma_1}{=} (c_1(\gamma_2z) + d_1)^k f(\gamma_2z) \\ &= (c_1 \cdot \frac{a_2z + b_2}{c_2z + d_2} + d_1)^k (c_2z + d_2)^k f(z) = ((c_1a_2 + c_2d_1)z + c_1b_2 + d_2d_1)^k f(z) \end{aligned}$$

Now,

$$\begin{aligned} f(z) &= f(\gamma_1(\gamma_1^{-1}z)) = (c_1(\gamma_1^{-1}z) + d_1)^k f(\gamma_1^{-1}z) \\ &= (c_1(\frac{d_1z - b_1}{-c_1z + a_1}) + d_1)^k f(\gamma_1^{-1}z) \\ &= (\frac{1}{-c_1z + a_1})^k f(\gamma_1^{-1}z) \end{aligned}$$

□

Examples (Non-constant modular forms).

(1) Eisenstein series: For $k \in \mathbb{Z}_{\geq 4}$ even,

$$\mathbb{G}_k := -\frac{B_k}{2k} + \sum_{n \geq 1} \sigma_{k-1}(n) \cdot q^n$$

where $q = e^{2\pi iz}$, $\sigma_s(n) = \sum_{d|n, d>0} d^s$, B_k is the k th Bernoulli number defined by the following Maclaurin series

$$\frac{x}{e^x - 1} = \sum_{k \geq 0} B_k \frac{x^k}{k!}$$

In particular

$$B_k \in \mathbb{Q}, \forall k \geq 0; B_k = 0 \text{ for all } k \geq 3 \text{ odd}$$

Normalaization:

$$E_k = \frac{G_k}{(-B_k/2k)}$$

(2) The Δ -function defined by

$$\Delta = \frac{E_4^3 - E_6^2}{1728} = \sum_{n \geq 1} \tau(n) q^n$$

where τ is Ramanyan function satisfying $\tau(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$. τ contains many conjectures which are essentially proved by modular forms.

Remark. Let M_k the set of modular forms of weight k then M_k is a $\mathbb{C}$ -vector space with finite dimensions. A basis consists of products of Eisenstein series and the Δ -functions.

Remark. If $f \in M_k, g \in M_{k'}$ then $fg \in M_{k+k'}$.

Remark (Exciting remark). Modular form is essential in the proof of Taniyama-Shimura-Weil Conjecture which inducts Fermat's Last theorem. To $f \in M_k$, we attach an L -function: $L(f, s) = \sum_{n \geq 1} a_n n^{-s}$ where $f = \sum_{n \geq 0} a_n q^n$. In fact $L(f, s)$ converges for $\text{Re}(s) \gg 0$ and $L(f, s)$ is like the Riemann zeta function. Taniyama-Shimura-Weil implies the L -function of an elliptic curve $\mathbb{Q}$ which coincides with the L -function of a modular form (for a subgroup of $SL_2(\mathbb{Z})$). And hence has a meromorphic continuity to $\mathbb{C}$.

1 The Modular Group

Definition 1.1. We call region $\mathcal{F} \subseteq \mathbb{H}$ a fundamental domain for the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$ if

- (i) For all $z \in \mathbb{H}$, there exists $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma z \in \mathcal{F}$.
- (ii) $\gamma z \neq z$ for all $z \in \text{Int}(\mathcal{F})$ and for all $\gamma \in SL_2(\mathbb{Z}) \setminus \{\pm I\}$.

Theorem 1.2. Let $\mathcal{F} = \{z \in \mathbb{H} \mid |\text{Re}(z)| \leq \frac{1}{2}, |z| \geq 1\}$ then the following statements hold:

- (1) $\mathcal{F}$ is a fundamental domain for the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$ i.e. if $z, z' \in \mathcal{F}, z \neq z'$ such that $z' = \gamma z$ for some $\gamma \in SL_2(\mathbb{Z})$ then either $|\text{Re}(z)| = \frac{1}{2}$ and $z' = T^\pm z$ or $z' = Sz$ on $|z| = 1$.
- (2) Let $\text{Stab}_{SL_2(\mathbb{Z})}(z) = \{\gamma \in SL_2(\mathbb{Z}) \mid \gamma z = z\}$. Then for all $z \in \mathcal{F}$, $\text{Stab}_{SL_2(\mathbb{Z})}(z) = \{\pm I\}$ except in the following cases:
 - (a) For $z = i$, $\text{Stab}_{SL_2(\mathbb{Z})}(i) = \langle S \rangle = \{\pm I, \pm S\}$.
 - (b) For $z = \rho = e^{\frac{2\pi i}{3}}$, $\text{Stab}_{SL_2(\mathbb{Z})}(\rho) = \langle ST \rangle = \{\pm I, \pm ST, \pm (ST)^2\}$.
 - (c) For $z = -\bar{\rho}$, $\text{Stab}_{SL_2(\mathbb{Z})}(-\bar{\rho}) = \langle TS \rangle = \{\pm I, \pm TS, \pm (TS)^2\}$.
- (3) $SL_2(\mathbb{Z}) = \langle S, T \rangle$.

Proof.

- (1) Let $z \in \mathbb{H}$. First we want to show that there exists $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma z \in \mathcal{F}$. Recall that

$$\text{Im}(\gamma z) = \frac{\text{Im}(z)}{|cz + d|^2}, \text{ where } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

We claim that $|cz + d|^2$ has a minimum as γ runs over $SL_2(\mathbb{Z})$ hence $\text{Im}(\gamma z)$ has a maximum.

Proof for the Claim. Let $z := x + iy$ where $x \in \mathbb{R}$ and $y \in \mathbb{R}_{>0}$. Let $M \in \mathbb{R}_{>0}$ then assume

$$|cz + d|^2 = (cx + d)^2 + c^2 y^2 < M \implies |c| < \frac{M}{y}$$

Thus, we only have finitely many $c \in \mathbb{Z}$ satisfying the inequality. In addition, we can also deduce that $|cx + d| < M$ so d can only take finitely many values in $\mathbb{Z}$. $\square$

Therefore $\text{Im}(\gamma z)$ has a maximum as γ runs over $SL_2(\mathbb{Z})$ so also as γ runs over $\langle S, T \rangle \subseteq SL_2(\mathbb{Z})$. Let $\gamma_0 \in \langle S, T \rangle$ be such an element i.e.

$$\text{Im}(\gamma_0 z) \geq \text{Im}(\gamma z), \text{ for all } \gamma \in \langle S, T \rangle.$$

Since $\gamma_0 \in \langle S, T \rangle$, $S\gamma_0 \in \langle S, T \rangle$. Therefore

$$\begin{aligned} \text{Im}(\gamma_0 z) &\geq \text{Im}((S\gamma_0)z) = \text{Im}(S(\gamma_0 z)) \\ &= \frac{\text{Im}(\gamma_0 z)}{|\gamma_0 z|^2} \end{aligned}$$

which is possible if and only if $|\gamma_0 z| \geq 1$. Now note that $\text{Im}(T^n \gamma_0 z) = \text{Im}(\gamma_0 z)$ for all $n \in \mathbb{Z}$ and there exists $N \in \mathbb{Z}$ such that $|\text{Re}(T^N \gamma_0 z)| \leq \frac{1}{2}$. Thus there exists some $\gamma = T^N \gamma_0 \in \langle S, T \rangle$ such that $\gamma z \in \mathcal{F}$.

Now we need to check that if $z' = \gamma z$ where $z, z' \in \mathcal{F}, \gamma \in SL_2(\mathbb{Z})$ then $z, z' \notin \text{Int}(\mathcal{F})$ unless $\gamma = \pm I$. Without loss of generality, let $\text{Im}(z') \geq \text{Im}(z)$. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then

$$\text{Im}(z') = \frac{\text{Im}(z)}{|cz + d|^2} \geq \text{Im}(z) \Leftrightarrow |cz + d| \leq 1$$

thus

$$|c| < \frac{1}{y} \leq \frac{1}{\sqrt{3}/2} < 2 \implies c \in \{0, 1, -1\} \text{ since } c \in \mathbb{Z}$$

For $c = 0$, $|cz + d| \leq 1 \implies d \in \{-1, 1\}$. Thus

$$\gamma = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = T^b \text{ or } \gamma = \begin{pmatrix} -1 & b \\ 0 & -1 \end{pmatrix} = -T^{-b}$$

Since $z, z' \in \mathcal{F}$, $b \in \{0, 1, -1\}$ thus either $\gamma = \pm I$ or $\gamma = \pm T$ and $\text{Re}(z) = -\frac{1}{2}$ or $\gamma = \pm I$ and $\text{Re}(z) = \frac{1}{2}$.

For $c = 1$, $|z + d| \leq 1$ implies that z is on the closed disk centred at $-d$ at radius 1 intersects $\mathcal{F}$ then either

$d = 0$ and z is on the arc joining ρ and $-\bar{\rho}$ or $d = 1$ and $z = \rho$ or $d = -1$ and $z = -\bar{\rho}$. For $d = 0$, $\gamma = \begin{pmatrix} a & -1 \\ 1 & 0 \end{pmatrix}$, then

$$\gamma z = \frac{az - 1}{z} = a - \frac{1}{z} = a - \bar{z} \in \mathcal{F}$$

$$\implies a = 0, \text{ so } \gamma = S \text{ or } a = 1, -\bar{z} = \rho, \gamma = TS \text{ or } a = -1, \gamma = (ST)^2.$$

we found TS fixes $-\bar{\rho}$ and hence ST fixes ρ since $(ST)^3 = -I$. In addition, S fixes i . In addition for $d = 1$, $z = \rho$, $\gamma = \begin{pmatrix} a & b \\ 1 & 1 \end{pmatrix} \implies a - b = 1$ so

$$z' = \gamma \rho = \frac{a\rho + a - 1}{\rho + 1} = a - \frac{1}{\rho + 1} = a + \rho$$

thus $a \in \{0, 1\}$. This means for $a = 0$, $\gamma = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = ST$ or $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. For $d = -1$ then $z = -\bar{\rho}$,

$$\gamma = \begin{pmatrix} a & b \\ 1 & -1 \end{pmatrix} \implies -a - b = 1 \implies b = -a - 1$$

$$\gamma(-\bar{\rho}) = \frac{-a\bar{\rho} - a - 1}{-1 - 1} = a - \bar{\rho}$$

Then $a = 0 \implies \gamma = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = (TS)^2$ and $a = -1 \implies \gamma = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix} = -ST^{-1}$.

(2) Covered before.

(3) Let $z_0 \in \text{Int}(\mathcal{F})$. Let $g \in SL_2(\mathbb{Z})$, let $z := gz_0 \in \mathbb{H}$. By the proof of (1), there exists $\gamma \in \langle S, T \rangle$ such that $\gamma z = (\gamma g)z_0 \in \mathcal{F}$ then $\gamma g = \pm I$ thus $g = \pm \gamma^{-1} \in \langle S, T \rangle$. □

Given a matrix $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, we want to write it as a product of S and T . Note that

$$T^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a + nc & b + nd \\ c & d \end{pmatrix}$$

while

$$S \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix}$$

If $|a| > |c|$ then apply the Division algorithm as follows $a = qc + r$, $0 \leq r < |c|$ then

$$T^{-q}\gamma = \begin{pmatrix} r & * \\ c & d \end{pmatrix}$$

Apply S , $ST^{-q}\gamma = \begin{pmatrix} -c & -d \\ r & * \end{pmatrix}$ until $r = 0$. If $|a| < |c|$, first apply S and then the above algorithm.

2 Modular form for $SL_2(\mathbb{Z})$

Definition 2.1. Let $k \in \mathbb{Z}$. A modular form of weight k for $SL_2(\mathbb{Z})$ is a function $f : \mathbb{H} \rightarrow \mathbb{C}$ given by a power series

$$f(z) = \sum_{n \geq 0} a_n q^n,$$

where $q = e^{2\pi iz}$, $a_n \in \mathbb{C}$, which converges for $|q| < 1$ and satisfies the identity $f(-\frac{1}{z}) = z^k f(z)$ for all $z \in \mathbb{H}$. A modular form is called cusp form if $a_0 = 0$, i.e. if f vanishes at $i\infty$.

Definition 2.2. Let $k \in \mathbb{Z}$. A modular function of weight k for $SL_2(\mathbb{Z})$ is a meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$, such that $f(z + 1) = f(z)$ and $f(-\frac{1}{z}) = z^k f(z)$, which is also meromorphic at $i\infty$.

Remark. A modular function f has a q -expansion of the form

$$f(z) = \sum_{n \geq -N} a_n q^n$$

for $\text{Im}(z)$ large enough, where $q = e^{2\pi iz}$, $a_n \in \mathbb{C}$, $N \in \mathbb{Z}_{\geq 0}$.

Definition 2.3. A lattice in $\mathbb{C}$ is a subgroup of $\mathbb{C}$ which is generated by a basis of $\mathbb{C}$ over $\mathbb{R}$. More precisely, pick $\omega_1, \omega_2 \in \mathbb{C}$ that are $\mathbb{R}$ -linearly independent, so $\text{Im}(\frac{\omega_1}{\omega_2}) \neq 0$. WLOG, we may assume $\text{Im}(\frac{\omega_1}{\omega_2}) > 0$. Then

$$\Lambda(\omega_1, \omega_2) := \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{a\omega_1 + b\omega_2 \mid a, b \in \mathbb{Z}\}$$

is a lattice in $\mathbb{C}$.

Example. The Gaussian integers $\Lambda(i, 1) = \mathbb{Z}i + \mathbb{Z} = \{ai + b \mid a, b \in \mathbb{Z}\}$ give us a lattice in $\mathbb{C}$ and remark that $\Lambda(i + n, 1) = \Lambda(i, 1)$ for all $n \in \mathbb{Z}$.

Let $Latt_{\mathbb{C}}$ be the set of lattices in $\mathbb{C}$ and let $M := \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in \mathbb{C}^\times, \text{Im}(\frac{\omega_1}{\omega_2}) > 0\}$. Now, if $(\omega_1, \omega_2) \in M$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, let $\omega'_1 = a\omega_1 + b\omega_2$, $\omega'_2 = c\omega_1 + d\omega_2$, so $\begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = \gamma \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}$. We claim that $(\omega'_1, \omega'_2) \in M$ and $\Lambda(\omega'_1, \omega'_2) = \Lambda(\omega_1, \omega_2)$.

Proof. We need to check that $\text{Im}(\frac{\omega'_1}{\omega'_2}) > 0$. But

$$\text{Im}(\frac{\omega'_1}{\omega'_2}) = \text{Im}(\frac{a\omega_1 + b\omega_2}{c\omega_1 + d\omega_2}) = \text{Im}(\gamma \frac{\omega_1}{\omega_2}) = \frac{\det(\gamma) \text{Im}(\frac{\omega_1}{\omega_2})}{|c\frac{\omega_1}{\omega_2} + d|^2} > 0$$

Since $\text{Im}(\frac{\omega_1}{\omega_2}) > 0$. Clearly $\Lambda(\omega'_1, \omega'_2) \subseteq \Lambda(\omega_1, \omega_2)$, since $\omega'_1, \omega'_2 \in \Lambda(\omega_1, \omega_2)$. And $\gamma^{-1} \begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}$ implies $\Lambda(\omega_1, \omega_2) \subseteq \Lambda(\omega'_1, \omega'_2)$. So $\Lambda(\omega'_1, \omega'_2) = \Lambda(\omega_1, \omega_2)$. $\square$

Lemma 2.4. Two elements of M define the same lattice in $\mathbb{C}$ if and only if they are related by a matrix in $SL_2(\mathbb{Z})$ as above.

Proof. We saw the condition is sufficient. To see it's necessary, suppose $\Lambda(\omega_1, \omega_2) = \Lambda(\omega'_1, \omega'_2)$. Then we must have $g, g' \in M_2(\mathbb{Z})$ such that $\begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = g \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}$ and $\begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} = g' \begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix}$. So $\begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = gg' \begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix}$. Since ω'_1, ω'_2 are $\mathbb{Z}$ -linearly independent, it must be the case that $gg' = I$, so $\det(g), \det(g') \in \{\pm 1\}$. But we see previously that we must have $\det(g), \det(g') > 0$ thus $\det(g) = \det(g') = 1$ i.e. $g, g' \in SL_2(\mathbb{Z})$. $\square$

Remark. We also have a surjective map $M \rightarrow \mathbb{H}$ such that $(\omega_1, \omega_2) \mapsto \frac{\omega_1}{\omega_2}$. But note that this is not injective. We encounter the action of $\mathbb{C}^\times$ on M . And we found the identification $M/\mathbb{C}^\times \longleftrightarrow \mathbb{H}$. Are the actions of Γ compatible. Note that $\gamma(\omega_1, \omega_2) = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ then

$$\frac{a\omega_1 + b\omega_2}{c\omega_1 + d\omega_2} = \gamma \left(\frac{\omega_1}{\omega_2} \right) \text{ where } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$$

Conversely: Let $z \in \mathbb{H}$, $\gamma z = \frac{az+b}{cz+d} \rightarrow (\frac{az+b}{cz+d}, 1) \sim (az+b, cz+d)$ in $M/\mathbb{C}^\times$. Then we must have

$$SL_2(\mathbb{Z}) \backslash \mathbb{H} \longleftrightarrow \mathbb{C}^\times \backslash Latt_{\mathbb{C}}$$

Remark. $\Lambda(\omega_1, \omega_2) = \omega_2 \Lambda(\frac{\omega_1}{\omega_2}, 1)$ such that $\text{Im}(\frac{\omega_1}{\omega_2}) > 0 \implies z := \frac{\omega_1}{\omega_2} \in \mathbb{H}$. Thus up to homothety, every lattice in $\mathbb{C}$ is of the form $\Lambda(z, 1)$ with $z \in \mathbb{H}$.

Remark. In general, for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$, $\Lambda(\gamma z, 1) \neq \Lambda(z, 1)$ but these two lattices are homothetic. Since

$$\begin{aligned} \Lambda\left(\frac{az+b}{cz+d}, 1\right) &= \frac{1}{cz+d} \Lambda(az+b, cz+d) \\ &= (cz+d)^{-1} \Lambda(z, 1) \end{aligned}$$

Thus, $\Lambda(\gamma z, 1) = (cz+d)^{-1} \Lambda(z, 1)$.

Definition 2.5. Let $k \in \mathbb{Z}$. A lattice of weight k is a function $F : Latt_{\mathbb{C}} \rightarrow \mathbb{C}$ such that

$$F(\lambda \Lambda) = \lambda^{-k} F(\Lambda), \quad \forall \lambda \in \mathbb{C}^\times, \forall \Lambda \in Latt_{\mathbb{C}}$$

Lemma 2.6. Let $k \in \mathbb{Z}$ then

- (1) If $F : Latt_{\mathbb{C}} \rightarrow \mathbb{C}$ is a lattice function of weight k then

$$f : \mathbb{H} \rightarrow \mathbb{C}$$

$$f(z) := F(\Lambda(z, 1))$$

satisfies the modularity condition with weight k .

- (2) Let $f : \mathbb{H} \rightarrow \mathbb{C}$ which satisfies the modularity condition with weight k then $F : Latt_{\mathbb{C}} \rightarrow \mathbb{C}$ satisfies

$$F(\Lambda(\omega_1, \omega_2)) = \omega_2^{-k} f\left(\frac{\omega_1}{\omega_2}\right)$$

is a lattice function of weight k for all $\omega_1, \omega_2 \in \mathbb{R}$ -linearly independent such that $\text{Im}(\frac{\omega_1}{\omega_2}) > 0$.

Proof.

(1) Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$. Then

$$\begin{aligned} f\left(\frac{az+b}{cz+d}\right) &= F\left(\Lambda\left(\frac{az+b}{cz+d}, 1\right)\right) \\ &= F((cz+d)^{-1}\Lambda(az+b, cz+d)) \\ &= (cz+d)^k \Lambda(z, 1) \end{aligned}$$

(2) Note that

$$\begin{aligned} F(\lambda\Lambda(\omega_1, \omega_2)) &= F(\Lambda(\lambda\omega_1, \lambda\omega_2)) \\ &= (\lambda\omega_2)^k f\left(\frac{\lambda\omega_1}{\lambda\omega_2}\right) \\ &= \lambda^{-k} \omega_2^{-k} f\left(\frac{\omega_1}{\omega_2}\right) \\ &= \lambda^{-k} F(\Lambda(\omega_1, \omega_2)) \quad \forall \lambda \in \mathbb{C}^\times \end{aligned}$$

So modular forms of weight k can be identifies with some lattice functions of weight k .

□

Definition 2.7 (Eisenstein Series). Let $k \in \mathbb{Z}_{>2}$ even. Fix $\Lambda \in \text{Latt}_{\mathbb{C}}$ with $\Lambda := \Lambda(\omega_1, \omega_2)$, let

$$G_k(\Lambda) := \sum_{\omega \in \Lambda, \omega \neq 0} \frac{1}{\omega^k} = \sum_{(m,n) \neq (0,0)} \frac{1}{(m\omega_1 + n\omega_2)^k}$$

We will show this series converges absolutely for $k > 2$ so the order of summation doesn't matter.

$$G_k(\Lambda) = \omega_2^{-k} \sum_{(m,n) \neq (0,0)} \frac{1}{(mz + n)^k}$$

where $z := \frac{\omega_1}{\omega_2} \in \mathbb{H}$.

Remark. Eisenstein Series is a lattice function since

$$\begin{aligned} G_k(\lambda\Lambda) &= \sum_{\omega \in \lambda\Lambda, \omega \neq 0} \frac{1}{\omega^k} = \sum_{\omega \in \Lambda} \frac{1}{(\lambda\omega)^k} \\ &= \lambda^{-k} G_k(\Lambda) \quad \forall \lambda \in \mathbb{C}^\times, \forall \Lambda \in \text{Latt}_{\mathbb{C}} \end{aligned}$$

so G_k is of weight k . Hence the corresponding function on $\mathbb{H}$, $G_k(z) := \sum_{(m,n) \neq (0,0)} \frac{1}{(mz+n)^k}$ will satisfy the modularity condition of weight k .

Remark. G_k makes sense for $k > 2$ odd but $G_k = 0$ in that case.

Proposition 2.8. The series defining $G_k(z)$, $z \in \mathbb{H}$ converges absolutely and uniformly on compact subset of $\mathbb{H}$ and hence defines a holomorphic function on $\mathbb{H}$.

Proof. We shall apply the following theorem to prove the proposition. We shall prove using the fact that $|mz+n|^2 \geq C(m^2+n^2)$ for some constant $C \in \mathbb{R} > 0$ then

$$\sum_{(m,n) \neq (0,0)} \left| \frac{1}{(mz+n)^k} \right| \leq \sum_{(m,n) \neq (0,0)} \frac{1}{(m^2+n^2)^{k/2}} \cdot \frac{1}{C^{k/2}}$$

and we are done. To be more precisely Let $z \in C_{r,s}$, $z = x + iy$ with $|x| \leq r, y \geq s$. For $m, n \in \mathbb{Z}$, $(m, n) \neq (0, 0)$ we have

$$|mz+n|^2 + (mx+n)^2 + m^2y^2 \geq ||mx| - |n||^2 + m^2s^2$$

If $\frac{|n|}{2} \geq |m| \cdot r \geq |mx|$ then $||mx| - |n|| = |n| - |mx| \geq \frac{|n|}{2}$. Thus, $|mz+n|^2 \geq \frac{|n|^2}{4} + m^2s^2 \geq \min\{\frac{1}{4}, s^2\}(m^2+n^2)$. If $\frac{|n|}{2} < |m|r$ then we are done. Thus we have $|mz+n|^2 \geq C(m^2+n^2)$ for some constant $C \in \mathbb{R} > 0$. Now we need to show that

$$\sum_{(m,n) \neq (0,0)} \frac{1}{(m^2+n^2)^{k/2}}$$

actually converges. For each $j \in \mathbb{Z}_{\geq 1}$, group together all the pairs $m, n \in \mathbb{Z}$ with the property that $\max\{m, n\} = j$ and we have $8j$ pairs. Thus we have

$$\sum_{m,n \in \mathbb{Z}} \frac{1}{(m^2+n^2)^{k/2}} \leq \sum_{j \geq 1} \frac{8j}{(j^2)^{k/2}} = \sum_{j \geq 1} \frac{8}{j^{k-1}}$$

which converges for $k > 2$.

□

Theorem 2.9 (Uniform limit theorem). Let $\Omega \neq \emptyset$ be an open, connected subset of $\mathbb{H}$. Let $(f_n)_{n \geq 1}$ be a sequence of holomorphic functions on Ω which converges uniformly on compact subsets of Ω . Then f is holomorphic on Ω .

Furthermore, the sequence of derivatives $\{f'_n\}$ converges uniformly to f' .

Theorem 2.10. Let $k \geq 4$ be an even integer. The Eisenstein series

$$G_k(z) := \sum_{(m,n) \neq (0,0)} \frac{1}{(mz+n)^k} = 2\zeta(k) + 2 \frac{(2\pi i)^k}{(k-1)!} \sum_{n \geq 1} \sigma_{k-1}(n) q^n$$

is a modular form of weight k for $SL_2(\mathbb{Z})$. In addition $G_k(i\infty) := 2\zeta(k)$ where ζ is the Riemann zeta function.

Proof. We shall use the identity that

$$\frac{\pi \cos(\pi z)}{\sin(\pi z)} = \frac{1}{z} + \sum_{n \geq 1} \frac{1}{z+n} + \frac{1}{z-n}.$$

The LHS of the above expression equals

$$\begin{aligned} \pi \cot(\pi z) &= \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{2\pi iz} + 1}{e^{2\pi iz} - 1} \\ &= \pi i \frac{q+1}{q-1} = -\pi i \left(1 + \frac{2q}{1-q}\right) \\ &= -\pi i - 2\pi i \sum_{n \geq 1} q^n \end{aligned}$$

for $z \in \mathbb{H}$, since in the case $|q| < 1$. Thus, we now have

$$\frac{1}{z} + \sum_{n \geq 1} \frac{1}{z+n} + \frac{1}{z-n} = -\pi i - 2\pi i \sum_{n \geq 1} q^n$$

Differentiating $(k-1)$ times, we get:

$$(-1)^{k-1} (k-1)! \left(\frac{1}{z^k} + \sum_{n \geq 1} \frac{1}{(z+n)^k} + \frac{1}{(z-n)^k} \right) = -(2\pi i)^k \sum_{n \geq 1} n^{k-1} q^n.$$

Thus for $k \geq 2$ even,

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^k} = \frac{(2\pi i)^k}{(k-1)!} \sum_{n \geq 1} n^{k-1} q^n.$$

Hence, for $k > 2$,

$$\begin{aligned} \sum_{m \geq 1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^k} &= \frac{(2\pi i)^k}{(k-1)!} \sum_{m \geq 1} \sum_{n \geq 1} n^{k-1} q^{mn} = \frac{(2\pi i)^k}{(k-1)!} \sum_{N \geq 1} q^N \sum_{d|N, d \geq 1} \\ &= \frac{(2\pi i)^k}{(k-1)!} \sum_{n \geq 1} \sigma_{k-1}(n) q^n \end{aligned}$$

Finally,

$$\begin{aligned} G_k(z) &= \sum_{m,n \neq (0,0)} \frac{1}{(mz+n)^k} = \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{1}{n^k} + \sum_{m \geq 1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^k} + \sum_{m \leq -1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^k} \\ &= 2 \sum_{n \geq 1} \frac{1}{n^k} + 2 \sum_{m \geq 1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^k} \end{aligned}$$

since k is even. So the above shows that

$$G_k(z) = 2\zeta(k) + 2 \frac{(2\pi i)^k}{(k-1)!} \sum_{n \geq 1} \sigma_{k-1}(n) q^n$$

□

Remark (Eisenstein Series of weight 2). Let

$$G_2(z) := \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^2} + 2\zeta(2)$$

this doesn't converge absolutely but it does converge. Then

$$G_2(z) = 2 \sum_{m \geq 1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^2} + 2\zeta(2)$$

Recall for $z \in \mathbb{H}$,

$$\begin{aligned} \pi \cot(\pi z) &= \frac{1}{z} + \sum_{n \geq 1} \frac{1}{z+n} + \frac{1}{z-n} \\ &= \pi i - 2\pi i \sum_{n \geq 0} q^n \quad q = e^{2\pi iz} \end{aligned}$$

Differentiation yields

$$-\frac{1}{z^2} - \sum_{n \geq 1} \frac{1}{(z+n)^2} + \frac{1}{(z-n)^2} = -(2\pi i)^2 \sum_{n \geq 1} nq^n$$

Then this implies

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^2} = (2\pi i)^2 \sum_{n \geq 1} nq^n$$

Replace z by mz gives

$$\sum_{m \geq 1} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^2} = (2\pi i)^2 \sum_{n \geq 1} \sigma_1(n)q^n$$

Thus,

$$G_2(k) = 2\zeta(2) + 2 \cdot (2\pi i)^2 \sum_{n \geq 1} \sigma_1(n)q^n$$

Let

$$E_2(z) := \frac{G_2(z)}{2\zeta(2)} = 1 - 24 \sum_{n \geq 1} \sigma_1(n)q^n$$

From the q -expansions, we see that $G_2(z+1) = G_2(z)$ and $E_2(z+1) = E_2(z)$. But these are not modular forms of weight 2.

Proposition 2.11. $G_2(-\frac{1}{z}) = z^2 G_2(z) - 2\pi i z$ and $E_2(-\frac{1}{z}) = z^2 E_2(z) - \frac{6iz}{\pi}$.

Proof. By definition,

$$\begin{aligned} G_2(-\frac{1}{z}) &= 2\zeta(2) + \sum_{m \geq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(m(-\frac{1}{z}) + n)^2} \\ &= 2\zeta(2) + \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{z^2}{(-m + nz)^2} \\ &= 2\zeta(2) + \sum_{n \neq 0} \sum_{m \in \mathbb{Z}} \frac{z^2}{(mz + n)^2} \end{aligned}$$

We want to relate $G_2(-\frac{1}{z})$ to $G_2(z)$ but we are not allowed to exchange the order of summation since we don't have absolute convergence. Trick: add to $G_2(-\frac{1}{z})$ and $G_2(z)$ specific series whose values we know so that the result are series that converge absolutely. Observe that

$$\sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{mz + n} - \frac{1}{mz + n + 1} = 0$$

But

$$\sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{mz + n} - \frac{1}{mz + n + 1} = -\frac{2\pi i}{z}$$

Subtract the first equation from $G_2(z)$ we get

$$\begin{aligned} G_2(z) &= 2\zeta(2) = \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(mz + n)^2} - \frac{1}{mz + n} + \frac{1}{mz + n + 1} \\ &= \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(mz + n)^2 (mz + n + 1)} \end{aligned}$$

which converges absolutely. Thus,

$$\begin{aligned} G_2(z) &= 2\zeta(2) + \sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{(mz + n)^2} - \left(\frac{1}{mz + n} - \frac{1}{mz + n + 1} \right) \\ &= 2\zeta(2) + \frac{1}{z^2} G_2(-\frac{1}{z}) - \left(\sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{mz + n} - \frac{1}{mz + n + 1} \right) - \frac{2\zeta(2)}{z^2} + \frac{2\zeta(2)}{z^2} - \frac{z^2 \cdot 2\zeta(2)}{z^2} \\ &= \frac{G_2(-\frac{1}{z})}{z^2} + \frac{2\pi i}{z} \end{aligned}$$

and we are done. □

Remark. In general

$$G_2\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} z\right) = (cz + d)^2 G_2(z) - 2\pi i c(cz + d)$$

for $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$.

3 Spaces of Modular Forms for $SL_2(\mathbb{Z})$

Proposition 3.1. For all $z \in \mathbb{H}$, $G_2(-\frac{1}{z}) = z^2 G_2(z) - 2\pi iz$ and $E_2(-\frac{1}{z}) = z^2 E_2(z) - \frac{6iz}{\pi}$.

Definition 3.2. $v_p(f)$ to be the order of the zero or pole of f at $p \in \mathbb{C}$.

Theorem 3.3. Let $k \in \mathbb{Z}$ and $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular function of weight k , $f \neq 0$. Then

$$v_{i\infty}(f) + \frac{v_i(f)}{2} + \frac{v_\rho(f)}{3} + \sum_{\bar{\rho} \in \Gamma \setminus \mathbb{H}} v_{\bar{\rho}}(f) = \frac{k}{12}$$

where $\sum'$ is the summation over all distinct classes $\neq \bar{i}, \bar{\rho} \in \Gamma \setminus \mathbb{H}$.

Remark. $\sum'$ is a finite sum and it is well-defined. f is meromorphic on $\mathbb{H} \cup \{i\infty\}$ then $\tilde{f}$ is meromorphic on D
 $\implies \exists r \in \mathbb{R}_{>0}$ such that $\tilde{f}$ has no zero on poles for $0 < |q| < r$

So f has no zeroes no poles on $\mathbb{H}$ for $\text{Im}(z) > \frac{1}{2\pi} \log(\frac{1}{r})$. All the poles of f on $\mathbb{H}$ are on the compact subset of $\mathbb{H}$

$$\mathcal{F}_r = \{z \in \mathcal{F} \mid \text{Im}(z) \leq \frac{1}{2\pi} \log\left(\frac{1}{r}\right), |\text{Re}(z)| \leq \frac{1}{2}\}$$

Proof. Choose an appropriate contour close to $\mathcal{F}_r$ and apply the Cauchy's argument principle.

(Case 1) Suppose f has not zeroes on poles on the boundary of $\mathcal{F}_r$ except possibly at $\rho, i, -\bar{\rho}$. So by assumptions zeroes and poles of f on h have exactly one representative inside C_r , except possibly the ones in $\bar{i}, \bar{\rho}, -\bar{\rho}$. By Cauchy's Argument principle

$$\begin{aligned} \frac{1}{2\pi i} \int_{C_r} \frac{f'(z)}{f(z)} dz &= \sum_{\bar{\rho} \in \Gamma \setminus \mathbb{H}} v_{\bar{\rho}}(f) \\ \frac{1}{2\pi i} \int_E^A \frac{f'(z)}{f(z)} dz &= \frac{1}{2\pi i} \int_{|q|=r} \frac{\tilde{f}'(q)}{\tilde{f}(q)} dq = -v_0(\tilde{f}) = -v_{i\infty}(f) \end{aligned}$$

Since $f(z+1) = f(z)$, it follows that

$$\int_A^B \frac{f'(z)}{f(z)} dz = \int_E^{D'} \frac{f'(z)}{f(z)} dz = - \int_{D'}^E \frac{f'(z)}{f(z)} dz$$

so they cancel out. On the circle of radius r around $\rho : C_\rho$:

$$\frac{1}{2\pi i} \int_{C_\rho} \frac{f'(z)}{f(z)} dz = -v_\rho(f)$$

The corresponding integral along the arc BB' is the fraction of the angle of the circle corresponding to the BB' times $-v_\rho(f)$. As $r \rightarrow 0$, $\rho B'$ tends to the tangent at ρ to the unit circle centred at the origin denoted by ℓ . So the angle corresponding to BB' , C_ρ is the angle between B_ρ and ℓ : $\frac{\pi}{3}$. Thus we must have

$$\frac{1}{2\pi i} \int_B^{B'} \frac{f'(z)}{f(z)} dz = \frac{\pi/3}{2\pi} \cdot (-v_\rho(f)) = -\frac{v_\rho(f)}{6}$$

Similarly,

$$\frac{1}{2\pi i} \int_D^{D'} \frac{f'(z)}{f(z)} dz = -\frac{v_{-\bar{\rho}}(f)}{6} = -\frac{v_\rho(f)}{6}$$

In addition

$$\begin{aligned} \frac{1}{2\pi i} \int_C^{C'} \frac{f'(z)}{f(z)} dz &= -\frac{v_i(f)}{2} \\ \frac{1}{2\pi i} \left(\int_{B'}^C \frac{f'(z)}{f(z)} dz + \int_{C'}^D \frac{f'(z)}{f(z)} dz \right) &= \frac{1}{2\pi i} \left(\int_{B'}^C \frac{f'(z)}{f(z)} dz - \int_{B'}^C \frac{f'(Sz)}{f(Sz)} d(Sz) \right) \end{aligned}$$

Since $f(Sz) = z^k f(z)$ we must have

$$\begin{aligned} \frac{f'(Sz)d(Sz)}{f(Sz)} &= kz^{k-1}f(z) + z^k f'(z) \\ \implies \frac{1}{2\pi i} \left(\int_{B'}^C \frac{f'(z)}{f(z)} dz - \int_{B'}^C \frac{f'(Sz)}{f(Sz)} d(Sz) \right) &= \frac{1}{2\pi i} \int_{B'}^C \left(-\frac{k}{z} dz \right) \end{aligned}$$

As $r \rightarrow 0$, $B'C$ corresponding to an angle of $\pi/6$ at the origin of the unit circle oriented anti-clockwise. Thus

$$\frac{1}{2\pi i} \int_{B'}^C \left(-\frac{k}{z} dz \right) = \frac{1}{12} \cdot \frac{k}{2\pi i} \int_{C(0,1)} \frac{1}{z} dz = \frac{k}{12}$$

Put all the integrals together we have the result required. Case 2 is just the contour C_r symmetrically for zeroes and poles on the boundary. $\square$

Corollary 3.4.

- (a) The Eisenstein's series E_4 has a simple zero $\rho = e^{2\pi i/3}$ and all other zeroes on $\mathbb{H}$ that are not equivalent to ρ under the action of Γ on $\mathbb{H}$.
- (b) The Eisenstein's series E_6 has a simple zero at i and all other zeroes on $\mathbb{H}$ that are not equivalent to i under the action of Γ on $\mathbb{H}$.
- (c) $\Delta = \frac{E_4^3 - E_6^2}{1728}$ has no zeroes on $\mathbb{H}$ and Δ has a simple zero at $i\infty$.

Proof.

- (a) In the valence formula, let $v(E_\rho) := \sum_{\bar{\rho} \in \Gamma \backslash \mathbb{H}} \nu_\rho(E_4)$ and

$$v_{i\infty}(E_4) + \frac{v_i(E_4)}{2} + \frac{v_\rho(E_4)}{3} + v(E_4) = \frac{4}{12} = \frac{1}{3}$$

$$\implies v_\rho(E_4) = 1, \text{ other terms are } 0$$

- (b) The argument is similar as above.
- (c) Same as before, let $v(E_4) := \sum_{\bar{\rho} \in \Gamma \backslash \mathbb{H}} \nu_\rho(\Delta)$ and we know $v_{i\infty}(\Delta) \geq 1$. Thus we must have other terms to be zero.

□

Corollary 3.5. Let M_k to be the set of modular forms of weight k for $SL_2(\mathbb{Z}) =: \Gamma$. Since $\mathbb{C}$ -linear combination of modular form of weight k are modular form of weight k , it's easy to see that M_k is a $\mathbb{C}$ -vector space. Let $S_k := S_k(\Gamma)$ be the subspace consisting of the cusp forms of weight k . Consider

$$\begin{aligned} \phi : M_k &\rightarrow \mathbb{C} \\ f &\mapsto i\infty \end{aligned}$$

Thus, $S_k = \ker(\phi)$ so it's a subspace of M_k . Note that

$$\dim_{\mathbb{C}}(M_k/S_k) \leq \dim_{\mathbb{C}}(\mathbb{C}) = 1$$

Note that for $k \geq 4$, k even, $\mathbb{C}E_k \subset M_k$ and $\mathbb{C}E_k \cap S_k = \{0\}$. Thus, $M_k = \mathbb{C}M_k \oplus S_k$.

Theorem 3.6.

- (a) $M_k = \{0\}$ for $k < 0$, $k = 2$ and $k \in \mathbb{Z}$ odd.
- (b) Multiplication by Δ defines an isomorphism from M_{k-12} to S_k . Hence by part(a) we get that $S_k = \{0\}$ for $k \leq 10$ so $M_4 = \mathbb{C}E_4$, $M_6 = \mathbb{C}E_6$, $M_8 = \mathbb{C}E_8$ and $M_{10} = \mathbb{C}E_{10}$.
- (c) $M_0 = \mathbb{C}$.
- (d) For $k \geq 0$ even,

$$\dim_{\mathbb{C}}(M_k) = \begin{cases} \lfloor \frac{k}{12} \rfloor & \text{if } k \equiv 2 \pmod{12} \\ \lfloor \frac{k}{12} \rfloor + 1 & \text{if } k \not\equiv 2 \pmod{12} \end{cases}$$

Proof.

- (a) $M_k = \{0\}$ for $k < 0$ by the valence formula since $LHS \geq 0$. Now let f be a modular form of weight 2, the valence formula gives

$$v_{i\infty}(f) + \frac{v_i(f)}{2} + \frac{v_\rho(f)}{3} + v(f) = \frac{1}{6}$$

which is impossible.

- (b) Let $f \in M_{k-12}(f)$, then $\Delta f \in M_k$. In addition, $(\Delta f)(i\infty) = 0$ i.e. it's a cusp form. We also have the inverse map Δ^{-1} given by

$$\begin{aligned} \Delta^{-1} : S_k &\rightarrow M_{k-12} \\ g &\mapsto \frac{g}{\Delta} \end{aligned}$$

$\Delta(z)$ has no zeroes for $z \in \mathbb{H}$ i.e. g/Δ is holomorphic on $\mathbb{H}$. Observe that $v_{i\infty}(\frac{g}{\Delta}) = v_{i\infty}(g) - v_{i\infty}(\Delta) = v_{i\infty}(g) - 1$. But since $g \in S_k$ so $v_{i\infty}(g) \geq 1$ in this case $v_{i\infty}(\frac{g}{\Delta}) \geq 0$. In conclusion $\frac{g}{\Delta} \in M_{k-12}$.

- (c) We know $\mathbb{C} \subset M_0$. From(b), we have $S_0 = \{0\}$ and

$$\begin{aligned} \dim_{\mathbb{C}}(M_0/S_0) &\leq 1 \\ \implies M_0 &= \mathbb{C} \end{aligned}$$

- (d) Use induction on k , the statements holds for all k even, $0 \leq k \leq 10$. Suppose $k \geq 12$ then

$$\begin{aligned} M_k &= \mathbb{C}E_k \oplus S_k \\ \implies \dim_{\mathbb{C}}(M_k) &= 1 + \dim_{\mathbb{C}}(M_{k-12}) \end{aligned}$$

Also when we replace k by $k + 12$ on the RHS of the formula, the RHS increases by 1 as well.

□

Examples.

- (1) $S_{12} \cong M_0 = \mathbb{C} \implies S_{12} = \mathbb{C}\Delta$. So a basis for M_{12} is $\{E_{12}, \Delta\}$ which also indicates the basis for S_{24} is $\{\Delta E_{12}, \Delta^2\}$.
- (2) $S_{14} \cong M_2 = \{0\}$.
- (3) $S_{16} \cong M_4 = \mathbb{C}E_4$. Thus, basis for $S_{16} = \{\Delta E_4\}$.
- (4) $\dim_{\mathbb{C}}(M_k) = 1$ then $E_8, E_4^2 \in M_k$ so $\exists c \in \mathbb{C}$ such that $E_8 = cE_4^2$. In fact $E_8 = E_4^2$. Similarly, $\dim_{\mathbb{C}}(M_{10}) = 1$ so $E_4E_6 = E_{10}$. $\dim_{\mathbb{C}}(M_{12}) = 2$ so $E_6^2, E_{12}, \Delta \in M_{12}$ we must have E_6^2, E_{12}, Δ linearly dependent.

Theorem 3.7 (Eisenstein Basis). Let $k \geq 0$ even, then the set $\{E_4^a E_6^b \mid 4a + 6b = k, a, b \in \mathbb{Z}_{\geq 0}\}$ is a basis for M_k .

Proof. $E_4^a E_6^b \in M_k$ for any $a, b \in \mathbb{Z}_{\geq 0}$ since $4a + 6b = k$. Thus their $\mathbb{C}$ -span $\subset M_k$. We'll do induction on k to show $M_k \subset$ the $\mathbb{C}$ -space of the set given. True for $k \leq 10$ from previous example. Suppose $k \geq 12$, let $f \in M_k$, we claim that $\exists c \in \mathbb{C}$ such that $f = cE_4^a E_6^b$ where $4a + 6b = k$, $a, b \in \mathbb{Z}_{\geq 0}$ is a cusp form. Also a, b exists by the induction hypothesis e.g. Let $a_0, b_0 \in \mathbb{Z}_{\geq 0}$ such that $4a_0 + 6b_0 = k - 4$ then $(a, b) = (a_0 + 1, b_0)$ will do the trick. Thus

$$f = cE_4^a E_6^b \in S_k \cong M_{k-12}$$

Again by the induction hypothesis $f \in \mathbb{C}$ -space

$$\{E_4^a E_6^b \mid a, b \in \mathbb{Z}_{\geq 0}, 4a + 6b = k\}.$$

To show this is a basis, we need to show linear independence. Proceed again by induction. True for $k \leq 10$ and even. Let $k \geq 12$, suppose

$$\sum_{4a+6b=k} c_{a,b} E_4^a E_6^b = 0$$

Notice then that each term must contain both E_4 and E_6 with a positive exponent. Suppose there exists a term containing $c_{k/4,0} E_4^{k/4}$ in the sum, all the others will contain E_6^b for some $b \in \mathbb{Z}_{k \geq 0}$. Plug in $z = i$: since $E_6(i) = 0 \implies c_{k/4,0} E_4^{k/4} = 0$. Since $E_4(i) \neq 0$. Now divide the relation by E_6 . Thus

$$\sum_{4a+6b=k} c_{a,b} E_4^a E_6^b = 0$$

gives a linear independence among conversing polynomials then $c_{a,b} = 0$ for all $a, b \in \mathbb{Z}_{\geq 0}$. $\square$

Corollary 3.8. Let $k \in \mathbb{Z}_{\geq 4}$, even. Let $b = 0$ if $k \equiv 0 \pmod{4}$ and $b = 1$ if $k \equiv 2 \pmod{4}$. The set

$$\{\Delta^j E_6^b E_4^{(k-12j-6b)/4} \mid 0 \leq j < \dim_{\mathbb{C}}(M_k)\}$$

is a $\mathbb{C}$ -basis for M_k . Furthermore, any $f \in M_k(\mathbb{Z})$ can be written as a $\mathbb{Z}$ -linear combination of elements in this basis.

Theorem 3.9. E_4 and E_6 are algebraically independent over $\mathbb{C}$. Hence $\mathbb{C}[E_4, E_6]$ is isomorphic to the polynomial ring $\mathbb{C}[X, Y]$.

Proof. Now suppose E_4 and E_6 are not algebraically independent over $\mathbb{C}$, i.e. that there exists a non-zero polynomial $P \in \mathbb{C}[X, Y]$ such that $P(E_4, E_6) = 0$. Since modular forms of different weights are linearly independent over $\mathbb{C}$, it must be the case that $P(E_4, E_6)$ is a sum of modular forms of the same weight k :

$$P(E_4, E_6) = \sum_{4i+6j=k} c_{i,j} E_4^i E_6^j,$$

with $c_{i,j} \in \mathbb{C}$. But since $\{E_4^a E_6^b \mid 4a + 6b = k; a, b \in \mathbb{Z}_{\geq 0}\}$ is a basis for M_k over $\mathbb{C}$, it must be the case that all the coefficients $c_{i,j}$ of P are equal to 0, and hence that $P = 0$, which is a contradiction. $\square$

Definition 3.10. Let $\eta : \mathbb{H} \rightarrow \mathbb{C}$ defined by

$$\eta(z) = e^{2\pi iz/24} \prod_{n \geq 1} (1 - q^n), \quad q = e^{2\pi iz}$$

The infinite product above converges absolutely and uniformly on compact subsets of $\mathbb{H}$, so it defines a holomorphic function, η on $\mathbb{H}$, which is called Dedekind's η -function. It is easy to see that $\eta(z+1) = e^{2\pi iz/24} \eta(z)$.

Proposition 3.11. For all $z \in \mathbb{H}$,

$$\eta\left(-\frac{1}{z}\right) = \sqrt{-iz} \eta(z)$$

where the branch of the sqrt is taken to have positive real part.

Proof. Take $d \log$:

$$\begin{aligned} \frac{d}{dz} \log(\eta(z)) &= \frac{2\pi i}{24} + \sum_{n \geq 1} \frac{-2\pi i q^n}{1 - q^n} = \frac{\pi i}{12} - 2\pi i \sum_{n \geq 1} n q^n \sum_{m \geq 0} (q^n)^m \\ &= \frac{\pi i}{12} - 2\pi i \sum_{n \geq 1} n \sum_{m \geq 1} q^{mn} = \frac{\pi i}{12} - 2\pi i \sum_{N \geq 1} q^N \sum_{d|N, d \geq 1} d \\ &= \frac{\pi i}{12} - 2\pi i \sum_{n \geq 1} \sigma_1(n) q^n = \frac{\pi i}{12} E_2(z) \end{aligned}$$

And

$$\begin{aligned} \frac{d}{dz} \log\left(\eta\left(-\frac{1}{z}\right)\right) &= \frac{d}{dz} \log(\eta(z))|_{z=-1/z} \cdot \frac{d}{dz}\left(-\frac{1}{z}\right) = \frac{\pi i}{12} E_2\left(-\frac{1}{z}\right) \cdot \frac{1}{z^2} \\ &= \frac{\pi i}{12z^2} (z^2 E_2(z) - \frac{6iz}{\pi}) = \frac{\pi i}{12} E_2(z) + \frac{1}{2z} \\ &= \frac{d}{dz} \log(\eta(z)) + \frac{d}{dz} \log(\sqrt{-iz}) = \frac{d}{dz} \log(\eta(z) \cdot \sqrt{-iz}) \end{aligned}$$

So we must have $\eta(-\frac{1}{z}) = c\sqrt{-iz}\eta(z)$, for some $c \in \mathbb{C}^\times$. Letting $z = i$, we get $\eta(i) = c\eta(i)$, so $c = 1$ since $\eta(i) \neq 0$, which concludes the proof. $\square$

Remark. Let $F(z) := \eta(z)^{24}$. Then $F(-\frac{1}{z}) = z^{12}F(z)$ and $F(z+1) = F(z)$ since $\eta(z+1) = \eta(z)$. F is holomorphic on $\mathbb{H}$ i.e. $F \in M_{12}$ and $F(z) = q \prod_{n \geq 1} (1 - q^n)^{24}$ with $q = e^{2\pi iz}$. Since $F(i\infty) = 0$ we must also have $F \in S_{12} = C\Delta$. F and Δ differs by a constant, which must be 1 from the q -expansion.

Definition 3.12. Let

$$\begin{aligned} j : \mathbb{H} &\rightarrow \mathbb{C} \\ z &\mapsto \frac{E_4^3(z)}{\Delta} \end{aligned}$$

Proposition 3.13. The j -invariant defines a bijection from $\Gamma \backslash \mathbb{H}$ to $\mathbb{C}$ is a bijection.

Proof. Let $j : \Gamma \backslash \mathbb{H} \rightarrow \mathbb{C}$, the map is well-defined because j is a modular function of weight 0. To show j is a bijection, we need to show that given $\lambda \in \mathbb{C}$, there exists $z \in \Gamma \backslash \mathbb{H}$ such that $f(z) = \lambda$. Let $f_\lambda := j - \lambda$, f_λ is a modular function of weight 0. f_λ has a simple pole at $i\infty$ and no poles on $\mathbb{H}$. So we can apply the Valence formula to f_λ

$$v_{i\infty}(f_\lambda) + \frac{v_i(f_\lambda)}{2} + \frac{v_\rho(f_\lambda)}{3} + \sum_{\bar{\rho} \in \Gamma \backslash \mathbb{H}} \nu_{\bar{\rho}}(f_\lambda) = \frac{0}{12} = 0$$

Thus we must have

$$\frac{v_i(f_\lambda)}{2} + \frac{v_\rho(f_\lambda)}{3} + \sum_{\bar{\rho} \in \Gamma \backslash \mathbb{H}} \nu_{\bar{\rho}}(f_\lambda) = 1$$

this has only non-negative solutions i.e. $(2, 0, 0)$, $(0, 3, 0)$, $(0, 0, 1)$. In all cases we have exactly one zero in $\Gamma \backslash \mathbb{H}$. e.g. $j(i) - 1728 = 0$, $j(\rho) = 0$. $\square$

Theorem 3.14. Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a meromorphic function. Then the following are equivalent:

- (1) f is a modular function of weight 0.
- (2) f is a quotient of two modular forms of the same weight;
- (3) f is a rational function of j , i.e. there exist $P, Q \in \mathbb{C}[X]$ such that

$$f(z) = \frac{P(j(z))}{Q(j(z))}$$

Proof. (3 $\implies$ 2) : Let $P(x) = a_p x^p + a_{p-1} x^{p-1} + \dots + a_0$ where $a_i \in \mathbb{C}$, $a_p \neq 0$. Then

$$\begin{aligned} P(j) &= a_p \frac{E_4^{3p}}{\Delta^{3p}} + a_{p-1} \frac{E_4^{3p-3}}{\Delta} + \dots + a_1 \frac{E_4^3}{\Delta} + a_0 \\ &= \frac{1}{\Delta^p} (a_p E_4^{3p} + a_{p-1} E_4^{3p-3} \Delta + \dots + a_0 \Delta^p) \end{aligned}$$

observe that all terms in fact have weight $12p$. Similarly, if $\deg(Q) = q$ then $Q(j) = \frac{1}{\Delta^q}$ times a modular form of weight $12q$. Thus,

$$\frac{P(j)}{Q(j)} = \frac{(\text{modular function of weight } 12p) \times \Delta^q}{\text{modular function of weight } 12q \times \Delta^p}$$

which is the quotient of two modular forms of the same weight.

(2 $\implies$ 1) : Clear.

(1 $\implies$ 3) : Let f be a modular function of weight 0 then f has only finitely many poles on $\Gamma \backslash \mathbb{H}$ say $a_1, \dots, a_N$.

Then if we look at

$$f \cdot \prod_{i=1}^N (j(z) - j(a_i))^{-v_{a_i}(f)}$$

which has no poles on $\mathbb{H}$. In addition, there exists some $M \in \mathbb{Z}$ such that

$$F := \Delta^M f \prod_{i=1}^N (j(z) - j(a_i))^{-v_{a_i}(f)}$$

has no pole at $i\infty$. Overall, F is a modular function of weight $12M$. If we see $\frac{F}{\Delta^M}$ is a rational function in j we are done. Note that

$$F = \sum_{4a+6b=12M; a,b \in \mathbb{Z}_{\geq 0}} c_{a,b} E_4^a E_6^b$$

Thus,

$$\frac{F}{\Delta^M} = \sum_{4a+6b=12M; a,b \in \mathbb{Z}_{\geq 0}} c_{a,b} \frac{E_4^a E_6^b}{\Delta^M}$$

Since $j = \frac{E_4^3}{\Delta}$ and $\frac{E_6^2}{\Delta} = j - 1728$ we must have

$$\sum_{4a+6b=12M; a,b \in \mathbb{Z}_{\geq 0}} c_{a,b} \left(\frac{E_4^3}{\Delta}\right)^{a/3} \left(\frac{E_6^2}{\Delta}\right)^{b/2}$$

Then this is equivalent to

$$\sum_{4a+6b=12M; a,b \in \mathbb{Z}_{\geq 0}} c_{a,b} j^{a/3} (j - 1728)^{b/2}$$

with exponents integer and we are done. □

4 Fourier Coefficients of Modular forms

Theorem 4.1 (Hecke Bounds). Let $f = \sum_{n \geq 0} a_n q^n \in M_k$ and $k \in \mathbb{Z}_{\geq 4}$ even.

- (1) If $f \in S_k$ then $\frac{|a_n|}{n^{k/2}}$ is bounded as $k \rightarrow \infty$ i.e. $a_n = O(n^{k/2})$.
- (2) If $f \in M_k \setminus S_k$ then $a_n = O(n^{k-1})$. In fact there exists $A, B \in \mathbb{R}_{>0}$ such that
$$An^{k-1} \leq |a_n| \leq Bn^{k-1}$$

Proof.

- (1) By Cauchy's Residue theorem,

$$a_n = \frac{1}{2\pi i} \oint_{C(0,r)} \frac{f(q)}{q^{n+1}} dq$$

Fix $y \in \mathbb{R}_{>0}$ and let x vary on $[0, 1]$. Then the integral becomes

$$\frac{1}{2\pi i} \oint_{C(0, e^{-2\pi y})} \frac{f(q)}{q^{n+1}} dq = \frac{1}{2\pi i} \int_0^1 \frac{f(q)}{q^{n+1}} 2\pi i q dx$$

i.e.

$$a_n = \int_0^1 \frac{f(q)}{q^n} dx$$

If we have a bound My for $|f(q)|$ that didn't depend on x then

$$a_n \leq \int_0^1 \frac{|f(q)|}{|q^n|} dx \leq Mye^{2\pi ny}, \quad \forall y \in \mathbb{R}_{>0}$$

Then pick y that minimize $Mye^{2\pi ny}$ to get a bound. We consider

$$\begin{aligned} \phi : \mathbb{H} &\rightarrow \mathbb{C} \\ z &\mapsto |f(z)|y^{k/2} \end{aligned}$$

which is continuously on $\mathbb{H}$ and $SL_2(\mathbb{Z})$ invariant. Note that $|\frac{f(q)}{q}| \rightarrow a_1$ as $q \rightarrow 0$ i.e. $|f(q)| = O(|q|) = O(e^{-2\pi y})$. So $\phi \rightarrow 0$ as $y \rightarrow +\infty$ which indicates it's bounded on the upper half plane. Therefore,

$$|a_n| \leq M \cdot e^{-2\pi ny} y^{-k/2}, \quad \forall y \in \mathbb{R}_{>0}$$

Take $y = 1/n$ we have

$$|a_n| \leq Mn^{k/2}$$

- (2) This is a consequence of (1), if $f \in M_k \setminus S_k$ then $f = \lambda E_k + g$ with $\lambda \neq 0$ and $g \in S_k$. Recall that

$$E_k = 1 - \frac{2k}{B_k} \sum_{n \geq 1} \sigma_{k-1}(n) q^n, \text{ and } g = \sum_{n \geq 1} b_n q^n, b_n = O(n^{k/2})$$

For $n \geq 1$,

$$a_n = -\frac{\lambda 2k}{B_k} \sigma_{k-1}(n) + b_n$$

It is enough to show that $\frac{\sigma_{k-1}(n)}{n^{k-1}}$ is bounded. Note that

$$1 \leq \frac{\sigma_{k-1}}{n^{k-1}} = \sum_{d|n, d>0} \left(\frac{d}{n}\right)^{k-1} = \sum_{d|n, d>0} \frac{1}{(n/d)^{k-1}}$$

which is equivalent to

$$\sum_{d'|n, d'>0} \frac{1}{(d')^{k-1}} \leq \sum_{d' \in \mathbb{Z}_{>0}} \frac{1}{(d')^{k-1}} = \zeta(k-1) < \infty$$

□

Remark. (1) is not the best we have: Deligne proved that $|a_n| = O(\sigma_0(n)n^{(k-1)/2})$.

5 Hecke Operators

Recall $\Delta = \frac{E_4^3 - E_6^2}{1728} = \sum_{n \geq 1} \tau(n)q^n = \prod_{n \geq 1} q(1 - q^n)^{24} \in S_{12}$ where $\tau : \mathbb{N} \rightarrow \mathbb{Z}$ is Ramanujan's τ -function. Around 1916, Ramanujan conjectured that

- (a) τ is multiplicative, i.e. $\tau(mn) = \tau(m)\tau(n)$ for all n, m prime.
- (b) For all $r \in \mathbb{N}$, $\tau(p^{r+1}) = \tau(p)\tau(p^r) - p^{11}\tau(p^{r-1})$.
- (c) The characteristic polynomial of the recurrence relation in (b), $x^2 - \tau(p)x + p^{11} = 0$, has negative discriminant, i.e. $(\tau(p))^2 - 4p^{11} < 0 \Leftrightarrow |\tau| < 2p^{11/2}$.

Definition 5.1. A sublattice of $\Lambda = \Lambda(\omega_1, \omega_2) \in \text{Latt}_{\mathbb{C}}$ is a lattice $\Lambda' \in \text{Latt}_{\mathbb{C}}$ which is a subgroup of Λ . The index of a sublattice Λ' of Λ is the size of the finite group Λ/Λ' and it is denoted by $[\Lambda : \Lambda']$.

Lemma 5.2. Let $\Lambda = \Lambda(\omega_1, \omega_2) \in \text{Latt}_{\mathbb{C}}$ and for $n \in \mathbb{N}$, let

$$S_n = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid a, b, d \in \mathbb{Z}, ad = n, a \geq 1, 0 \leq b < d \right\}$$

Then the map $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \rightarrow \Lambda(a\omega_1 + b\omega_2, d\omega_2)$ defines a bijection from S_n to the set of sublattices of Λ of index b .

Proof. Firstly, let's check that the map above does indeed yield sublattices of index n in Λ : Let $\Lambda' := \Lambda(a\omega_1 + b\omega_2, d\omega_2) \subset \Lambda$ and $\lambda'' := \Lambda(d(a\omega_1 + b\omega_2), d\omega_2)$. Then $[\Lambda', \Lambda''] = d$. Now, $\Lambda'' := \Lambda(d(a\omega_1 + b\omega_2), d\omega_2) = \Lambda(da\omega_1, d\omega_2)$, hence $[\Lambda : \Lambda''] = d^2a = dn$. Since $\Lambda'' \subset \Lambda' \subset \Lambda$ and $[\Lambda' : \Lambda''] = d$, it follows that $[\Lambda : \Lambda'] = da = n$ as required.

Also, note that the sublattices $\Lambda(a\omega_1 + b\omega_2, d\omega_2)$ with $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in S_n$ are distinct. These two sublattices are equal if and only if there exists $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$ where $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} \in S_n$. And it is easy to see that in this case $\gamma = I_2$.

Conversely, let $\Lambda' \subset \Lambda$, $[\Lambda : \Lambda'] = n$. Let $H_1 = \Lambda/(\Lambda' + \mathbb{Z}\omega_2)$ and $H_2 = (\Lambda' + \mathbb{Z}\omega_2)/\Lambda'$ and consider the surjective group homomorphism

$$\begin{aligned} \phi : \Lambda/\Lambda' &\rightarrow H_1 \\ \lambda + \Lambda' &\mapsto \lambda + (\Lambda' + \mathbb{Z}\omega_2) \end{aligned}$$

Since $\ker(\phi) = H_2$, we must have $n = |\Lambda/\Lambda'| = |H_1| * |H_2|$. Let $|H_1| = a$ and $|H_2| = d$. $|H_2| = d \implies d\omega_2 \in \Lambda'$ and $|H_1| = a \implies a\omega_1 \in \Lambda' + \mathbb{Z}\omega_2$. The above implies $\exists b \in \mathbb{Z}$ such that $a\omega_1 + b\omega_2 \in \Lambda'$ and we may choose $0 \leq b < d$ since $d\omega_2 \in \Lambda'$. □

Example. The sublattices of Λ of index $p \in \mathbb{Z}_{\geq 0}$ primes correspond to the matrices

$$\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & p \end{pmatrix}, \quad \begin{pmatrix} 1 & p-1 \\ 0 & p \end{pmatrix}$$

Definition 5.3. Let $\mathcal{L} :=$ the free abelian group on $\text{Latt}_{\mathbb{C}}$. For $n \in \mathbb{N}$, we define the following $\mathbb{Z}$ -linear operator

$$\begin{aligned} T(n) : \mathcal{L} &\rightarrow \mathcal{L} \\ T(n)\Lambda &= \sum_{[\Lambda : \Lambda'] = n} \Lambda' \end{aligned}$$

i.e. for $\Lambda \in \mathcal{L}$, $T(n)\Lambda$ is equal to the finite sum in $\mathcal{L}$ of the sublattices of Λ of index n . For $\lambda \in \mathbb{C}^\times$, we let R_λ be the $\mathbb{Z}$ -linear operator

$$\begin{aligned} R_\lambda : \mathcal{L} &\rightarrow \mathcal{L} \\ R_\lambda(\Lambda) &\rightarrow \lambda\Lambda. \end{aligned}$$

Proposition 5.4.

- (1) $\mathbb{R}_\lambda \circ R_{\lambda'} = R_{\lambda\lambda'}$ for all $\lambda, \lambda' \in \mathbb{C}^\times$.
- (2) $R_\lambda \circ T(n) = T(n) \circ R_\lambda$ for all $\lambda \in \mathbb{C}^\times, n \in \mathbb{N}$.
- (3) $T(mn) = T(m) \circ T(n)$ for all m, n coprime.
- (4) $T(p^n) \circ T(p) = T(p^{n+1}) + pR_p \circ T(p^{n-1})$ for all $n \in \mathbb{N}$ and $p \in \mathbb{N}$ prime.

Proof.

- (1) Clear.
- (2) Let $\Lambda \in \text{Latt}_{\mathbb{C}}$,

$$\begin{aligned} (R_\lambda \circ T(n))(\Lambda) &= R_\lambda(T(n)(\Lambda)) \\ &= R_\lambda\left(\sum_{[\Lambda:\Lambda']=n} \Lambda'\right) \\ &= \sum_{\Lambda:\Lambda'=n} \lambda\Lambda' \end{aligned}$$

On the other hand,

$$\begin{aligned} (T(n) \circ R_\lambda)(\Lambda) &= T(n)(\lambda\Lambda) \\ &= \sum_{[\lambda\Lambda:\Lambda'']=n} \Lambda'' \end{aligned}$$

Note that $\Lambda'' \subset \lambda\Lambda \implies \Lambda'' = \lambda\Lambda'$ for a sublattice Λ' of Λ . And $[\lambda\Lambda:\lambda\Lambda'] = [\Lambda:\Lambda']$ thus we get the equality there.

- (3) By definition

$$\begin{aligned} T(mn)(\Lambda) &= \sum_{[\Lambda:\Lambda'']=nm} \Lambda'' \\ (T(m) \circ T(n))(\Lambda) &= T(m)(T(n)\Lambda) \\ &= T(m)\left(\sum_{[\Lambda,\Lambda']=n} \Lambda'\right) \\ &= \sum_{[\Lambda:\Lambda']=n} T(m)\Lambda' \\ &= \sum_{[\Lambda:\Lambda']=n} \sum_{[\Lambda':\Lambda'']=m} \Lambda'' \end{aligned}$$

In the second we are summing over pairs (Λ', Λ'') such that

$$\Lambda'' \subset \Lambda' \subset \Lambda$$

For each $\Lambda'' \subset \Lambda$ of index mn , there exists unique $\Lambda' \subset \Lambda$ of index n containing Λ'' where $\text{hcf}(m, n) = 1$. There is one-to-one correspondence between such sublattices Λ' and subgroups of Λ/Λ'' of order m . $|\Lambda/\Lambda''| = mn$ and $\text{hcf}(m, n) = 1$ so Λ/Λ'' has a unique subgroup of order m .

- (4) By definition

$$\begin{aligned} (T(p^n) \circ T(p))(\Lambda) &= T(p^n)\left(\sum_{[\Lambda:\Lambda']=p} \Lambda'\right) \\ &= \sum_{[\Lambda:\Lambda']=p} \sum_{[\Lambda':\Lambda'']=p^n} \Lambda'' \end{aligned}$$

so sum over pairs (Λ', Λ'') such that $\Lambda'' \subset \Lambda' \subset \Lambda$ so $[\Lambda:\Lambda''] = p^{n+1}$.

In addition,

$$\begin{aligned} (R_p \circ T(p^{n-1}))(\Lambda) &= \sum_{[\Lambda:\Lambda']=p^{n-1}} p\Lambda' \\ &= \sum_{\substack{\Lambda'' \subset p\Lambda \\ [\Lambda:\Lambda'']=p^{n+1}}} \Lambda'' \end{aligned}$$

In addition,

$$T(p^{n+1})(\Lambda) = \sum_{[\Lambda:\Lambda'']=p^{n+1}} \Lambda''$$

All sums involve sublattices Λ'' of Λ of index p^{n+1} . In addition, in $\sum_{[\Lambda:\Lambda'']=p^{n+1}} \Lambda''$, all such Λ'' appear exactly once. Let $\Lambda'' \subset \Lambda$ be such a lattice. We need to show that the number of times Λ'' appears on LHS is equal to the number of times Λ'' appears on the RHS. Let a be the number of times Λ'' appear in the second equation and b be the number of times Λ'' appear in the first equation. We want to show $b = 1 + ap$.

Looking at the second equation, we have 2 cases, Case 1:

$$\Lambda'' \not\subseteq p\Lambda \implies a = 0$$

i.e. we must show $b = 1$. Thus, there exists a unique $\Lambda' \subset \Lambda$, $[\Lambda : \Lambda'] = p$ such that $\Lambda'' \subset \Lambda'$. We have one such Λ such that $\Lambda' = \Lambda'' + p\Lambda$ then $p\Lambda \subsetneq \Lambda'' + p\Lambda \subsetneq \Lambda$. $[p\Lambda : \Lambda] = p^2$ so $[\Lambda : \Lambda'' + p\Lambda] = p$. Suppose there exists $\Lambda', \tilde{\Lambda} \subset \Lambda$ of index p such that

$$\Lambda' \neq \tilde{\Lambda} \text{ and } \Lambda'' \subset \Lambda', \Lambda'' \subset \tilde{\Lambda}$$

Then $\Lambda'' \subset \Lambda' \cap \tilde{\Lambda}$. Note that

$$\begin{aligned} |\Lambda/\Lambda'| &= |\Lambda/\tilde{\Lambda}| \implies p\Lambda \subset \Lambda' \cap \tilde{\Lambda} \\ &\implies p\Lambda \subset \Lambda' \cap \tilde{\Lambda} \subsetneq \Lambda' \subsetneq \Lambda \end{aligned}$$

Thus, $p\Lambda = \Lambda' \cap \tilde{\Lambda}$. However, since $\Lambda'' \not\subseteq p\Lambda$, we reach a contradiction here since $\Lambda'' \subset \Lambda' \cap \tilde{\Lambda}$.

Case 2: $\Lambda'' \subseteq p\Lambda$ then $a = 1$. We want to show that $b = 1 + p$ i.e. there are exactly $p + 1$ sublattices Λ' of Λ , $[\Lambda : \Lambda'] = p$ such that $\Lambda'' \subseteq \Lambda'$. Now, $[\Lambda : \Lambda'] = p \implies p\Lambda \subseteq \Lambda'$ so $\Lambda'' \subset \Lambda'$. So all we need to do is count the number of sublattices of index p in Λ . Use the Matrix Lemma, $|S_p| = p + 1$ and we are done. $\square$

Corollary 5.5.

- (1) $T(p^n)$ is a polynomial in $T(p)$ and R_p for all $n \in \mathbb{Z}_{\geq 0}$ and all $p \in \mathbb{Z}_{\geq 0}$ prime.
- (2) $T(n)$ and $T(m)$ commute for all integers $n, m \in \mathbb{Z}_{\geq 0}$ i.e.

$$T(n)T(m) = T(m)T(n)$$

In fact

$$T(n)T(m) = T(m)T(n) = \sum_{a|hcf(n,m)} a R_a T\left(\frac{mn}{a^2}\right)$$

Proof.

- (1) By induction.
- (2) Exercise sheet 2.

$\square$

Definition 5.6. Given $F : Latt_{\mathbb{C}} \rightarrow \mathbb{C}$, we can extend it by linearity to $F : \mathcal{L} \rightarrow \mathbb{C}$ where

$$\mathcal{L} := \left\{ \sum_{\Lambda \in Latt_{\mathbb{C}}} n_{\Lambda} \Lambda \mid n_{\Lambda} \in \mathbb{Z} \right\}$$

Then we have

$$F\left(\sum_{\Lambda \in Latt_{\mathbb{C}}} n_{\Lambda} \Lambda\right) = \sum_{\Lambda \in Latt_{\mathbb{C}}} n_{\Lambda} F(\Lambda)$$

For $\lambda \in \mathbb{C}^{\times}$ we let

$$(R_{\lambda}F)(\Lambda) = F(\lambda\Lambda), \quad \forall \Lambda \in Latt_{\mathbb{C}}$$

Remark. If F is a lattice function of weight k , then

$$R_{\lambda}F = \lambda^{-k}F$$

Definition 5.7. We will say $F : \mathcal{L} \rightarrow \mathbb{C}$ is of weight k if $R_{\lambda}F = \lambda^{-k}F$ for all $\lambda \in \mathbb{C}$. For $F : \mathcal{L} \rightarrow \mathbb{C}$, $n \in \mathbb{Z}_{\geq 0}$ of weight k then

$$(T(n)F)(\Lambda) := n^{k-1}F(T(n)\Lambda) = n^{k-1} \sum_{[\Lambda:\Lambda']=n} F(\Lambda')$$

Remark. $T(n)F$ is also of weight k . Let $\lambda \in \mathbb{C}^{\times}$ then

$$\begin{aligned} R_{\lambda}(T(n)F)(\Lambda) &= R_{\lambda}(n^{k-1} \sum_{[\Lambda:\Lambda']=n} F(\Lambda')) \\ &= n^{k-1} \sum_{[\Lambda:\Lambda']=n} F(\lambda\Lambda') \\ &= n^{k-1} \sum_{[\Lambda:\Lambda']=n} \lambda^{-k} F(\Lambda') \\ &= \lambda^{-k} (T(n)F)(\Lambda) \end{aligned}$$

Lemma 5.8.

- (1) Let $F : \mathcal{L} \rightarrow \mathbb{C}$ of weight k , Then

$$T(m) \circ T(n)F = T(mn)F, \quad \forall m, n \in \mathbb{Z} \text{ coprime}$$

- (2) $T(p^n)T(p)F = T(p^{n+1})F + p^{k-1}T(p^{n-1})F$.

(3) $T(m)$ and $T(n)$ commute for all $m, n \in \mathbb{Z}_{\geq 0}$. In particular

$$T(m)T(n)F = \sum_{a|hcf(m,n)} a^{k-1} T\left(\frac{mn}{a^2}\right)F$$

Proof.

(1)

(2)

$$\begin{aligned} (T(p^n(T(p)F))(\Lambda) &= T(p^n)(p^{k-1} \sum_{[\Lambda:\Lambda']=p} F(\Lambda')) \\ &= T(p^n)(p^{k-1} F(T(p)\Lambda)) \\ &= (p^n)^{k-1} p^{k-1} F(T(p^n)T(p)\Lambda) \\ &= (p^{n+1})^{k-1} (F(T(p^{n+1})\Lambda) + F(pR_p T(p^{n-1})\Lambda)) \\ &= (T(p^{n+1})F)(\Lambda) + (p^{n+1})^{k-1} p F(R_p T(p^{n-1})(\Lambda)) \\ &= (T(p^{n+1})F)(\Lambda) + (p^{n+1})^{k-1} p p^{-k} F(T(p^{n-1})(\Lambda)) \\ &= (T(p^{n+1})F)(\Lambda) + p^{k-1} (T(p^{n-1}F)(\Lambda)) \end{aligned}$$

(3)

□

Definition 5.9. Let f be a modular function of weight k for $SL_2(\mathbb{Z})$. Let $f : \mathcal{L} \rightarrow \mathbb{C}$ be the associated lattice function:

$$F(\Lambda(z, 1)) = f(z)$$

In general,

$$F(\Lambda(\omega_1, \omega_2)) = \omega_2^{-1} f\left(\frac{\omega_1}{\omega_2}\right), \quad \text{Im}\left(\frac{\omega_1}{\omega_2}\right) > 0$$

For $n \in \mathbb{Z}_{>0}$,

$$\begin{aligned} (T(n)f)(z) &= (T(n)F)(\Lambda(z, 1)) \\ &= n^{k-1} \sum_{[\Lambda:\Lambda']=n} F(\Lambda') \end{aligned}$$

which can be described explicitly by Matrix Lemma. This is equivalent to

$$n^{k-1} \sum_{\gamma \in S_n} F(\Lambda(az + b, d)) = n^{k-1} \sum_{\gamma \in S_n} F(d\Lambda\left(\frac{az+b}{d}, 1\right)) = n^{k-1} \sum_{\gamma \in S_n} d^{-k} f\left(\frac{az+b}{d}\right)$$

where $\gamma = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$. Then if f is meromorphic/holomorphic on $\mathbb{H}$ then $T(n)f$ is also meromorphic/holomorphic on $\mathbb{H}$. If f has weight k , F has weight k so $T(n)f$ has weight k as well so it satisfies the modularity condition. We need to check the behaviour of $T(n)f$ at $i\infty$ to see if $T(n)f$ is also a modular function.

Lemma 5.10. Let f be a modular form of weight k for $SL_2(\mathbb{Z})$,

$$f(z) = \sum_{m \in \mathbb{Z}} c(m) q^m$$

where all but finitely many $c(m)$'s with $m < 0$ are equal to zero. Then

$$T(n)f = \sum_{m \in \mathbb{Z}} \eta(m) q^m$$

where $\eta(m) = \sum_{a|hcf(m,n)} a^{k-1} c\left(\frac{mn}{a^2}\right)$.

Proof. Note that

$$\begin{aligned} n^{k-1} \sum_{\gamma \in S_n} d^{-k} f\left(\frac{az+b}{d}\right) &= n^{k-1} \sum_{\substack{ad=n \\ a,d>0 \\ 0 \leq b < d}} d^{-k} \sum_{m \in \mathbb{Z}} c(m) e^{2\pi i m \left(\frac{az+b}{d}\right)} \\ &= n^{k-1} \sum_{m \in \mathbb{Z}} c(m) \sum_{\substack{ad=n \\ a,d>0}} d^{-k} e^{2\pi i m a z / d} \sum_{0 \leq b < d} e^{2\pi i m b / d} \end{aligned}$$

But since

$$\sum_{0 \leq b < d} e^{2\pi i m b / d} = \begin{cases} d & \text{if } d|m \\ \frac{(e^{2\pi i m / d})^d - 1}{e^{2\pi i m / d} - 1} = 0 & \text{otherwise} \end{cases}$$

Let $m' = m/d$ then it's equivalent to

$$n^{k-1} \sum_{m' \in \mathbb{Z}} \sum_{a|n} c\left(\frac{m'n}{a}\right) e^{2\pi i m' a z} \left(\frac{n}{a}\right)^{1-k} = \sum_{m \in \mathbb{Z}} e^{2\pi i m z} \sum_{a|hcf(m,n)} a^{k-1} c\left(\frac{mn}{a^2}\right)$$

□

Corollary 5.11. Let $n \in \mathbb{N}$. If f is a modular function, a modular form, respectively a cusp form of weight k for $SL_2(\mathbb{Z})$, then so is $T(n)f$.

Proof. Left to check the behaviour at $i\infty$ with the same notation as before

$$\gamma(m) = \sum_{a|hcf(m,n), a>0} a^{k-1} c\left(\frac{mn}{a^2}\right)$$

So if, for example, f is cusp form $c(m) = 0$ for all $m < 0$ so $\gamma(m) = 0$ for all $m < 0$. If f is cusp form then $c(0) = 0$ $\gamma(0) = 0$. Thus, we can extend the $T(n)$'s by linearity to linear operators on M_k . These are called the Hecke operators. Notice that the $T(n)$'s preserve S_k . □

Lemma 5.12. The relations satisfied by the $T(n)$'s on lattice functions will also hold on modular forms which is equivalent to previous Lemma:

- (1) $T(mn)f = T(m)T(n)f$ for all $m, n \in \mathbb{Z}_{>0}$, $(m, n) = 1$.
- (2) For $p \in \mathbb{Z}_{>0}$ prime, $k \in \mathbb{Z}_{>0}$, $T(p^{k+1})f = T(p^n)T(p)f - p^{k-1}T(p^{k-1})f$.
- (3) $T(m)T(n)f = T(n)T(m)f$ for all $m, n \in \mathbb{Z}$.

SO $T(p^k)$ is a polynomial in $T(p)$.

Definition 5.13. A non-zero modular form

$$f(z) = \sum_{m \geq 0} c(m)q^m$$

of weight $k \in \mathbb{Z}_{>2}$ for $SL_2(\mathbb{Z})$ is called a Hecke eigenform if for each $n \in \mathbb{N}$, there exists $\lambda(n) \in \mathbb{C}$ such that

$$T(n)f = \lambda(n)f$$

If $c(1) = 1$, then f is called a normalized Hecke eigenform.

Remark. It is enough to check that f is an eigenvector for all $T(p)$ with $p \in \mathbb{Z}_{>0}$, to prove that f is an eigenform.

Examples.

- (1) $S_{12} = \mathbb{C}\Delta$ hence Δ must be a Hecke eigenform. In fact, Δ is a normalized Hecke eigenform.
- (2) $M_4 = \mathbb{C}E_4 = \mathbb{C}\mathbb{G}_4$, so $E_4, \mathbb{G}_4$ are Hecke eigenforms.
- (3) $M_6 = \mathbb{C}E_6 = \mathbb{C}\mathbb{G}_6$ so $E_6, \mathbb{G}_6$ are Hecke eigenforms.
- (4) For $k \in \{16, 18, 20, 22, 26\}$, $S_k = \mathbb{C}E_{k-12}\Delta$, so $E_{k-12}\Delta$ is a Hecke eigenform for these weights.

Proposition 5.14. Let $k \in \mathbb{Z}_{>2}$ and $f(z) = \sum_{m \geq 0} c(m)q^m \in M_k$ be a Hecke eigenform: for $n \in \mathbb{N}$, $T(n)f = \lambda(n)f$, $\lambda(n) \in \mathbb{C}$. Then

- (1) The coefficient of q in the q -expansion of $T(n)f$, $\lambda(n)c(1)$ is equal to $c(n)$. So if f is a normalized Hecke eigenform, then $\lambda(n) = c(n)$, i.e. the Hecke eigenvalues coincide with the Fourier coefficients of f .
- (2) $c(1) \neq 0$.
- (3) If f is a normalized Hecke eigenform, then:
 - (i) $c(m)c(n) = c(mn)$, $\forall m, n \in \mathbb{N}$ coprime.
 - (ii) $c(p^n)c(p) = c(p^{n+1}) + p^{k-1}c(p^{n-1})$ for all $n \in \mathbb{N}$, $p \in \mathbb{N}$ prime.
 - (iii) $c(m)c(n) = \sum_{a|gcd(m,n)} a^{k-1} c\left(\frac{mn}{a^2}\right)$ for all $m, n \in \mathbb{N}$.

Proof.

- (1) We want to show $\lambda(n)c(1) = c(n)$ for all $n \in \mathbb{Z}_{>0}$. By definition,

$$f = \sum_{m \geq 0} c(m)q^m$$

Then

$$T(n)f = \sum_{m \geq 0} \gamma(m)q^m, \quad \gamma(m) = \sum_{\substack{a|hcf(m,n) \\ a>0}} a^{k-1} c\left(\frac{mn}{a^2}\right)$$

Note that $T(n)f = \lambda(n)f$. We shall compare the Fourier coefficients to q on both sides then

$$\gamma(1) = \sum_{\substack{a|hcf(1,n) \\ a>0}} a^{k-1} c\left(\frac{n}{a^2}\right) = c(n)$$

On the RHS, it's $\lambda(n)c(1)$. Thus, $c(n) = \lambda(n)c(1)$.

(2) $c(1) \neq 0$ otherwise by (1), $f = 0$.

(3) If $c(1) = 1$, then by (1), $\lambda(n) = c(n)$. The eigenvalues $\lambda(n)$ satisfy the same relations as the $T(n)$'s by previous Lemma. □

Corollary 5.15. Two modular forms of weight $k > 2$ for $SL_2(\mathbb{Z})$ which are normalized Hecke eigenforms with the same Hecke eigenvalues must coincide.

Remark. This proves Ramanujan's conjecture for the τ -function. Since $\Delta = \sum_{m \geq 1} \tau(m)q^m$ is a normalized eigenform.

Proposition 5.16. The eigenvalues of the Hecke operators $T(n)$, $n \in \mathbb{Z}_{>0}$ acting on M_k are algebraic integers.

Proof. Think of the characteristic polynomial associated to $T(n)$. We need to fix a basis of M_k , recall we have a basis of M_k consisting of integer Fourier coefficients. Furthermore, every modular form with integer Fourier coefficient can be written as $\mathbb{Z}$ -linear combination of elements in this basis. Pick such a basis. Since $T(n)$ applied to a modular form of integer coefficients gives another such modular form. Thus, we get that matrix associated to $T(n)$ has integer coefficient. Therefore, the characteristic polynomial associated to $T(n)$ has integer coefficients. □

Proposition 5.17. G_k for $k \geq 4$ even is a Hecke eigenform and

$$T(n)G_k = \sigma_{k-1}(n)G_k, \quad \forall n \in \mathbb{Z}_{>0}$$

Proof. Recall that it's enough to show that G_k is an eigenvector for all $T(p)$, $p \in \mathbb{Z}_{>0}$ prime i.e. they are the Fourier coefficients of $\mathbb{G}_k$.

Recall $G_k(z) = G_k(\Lambda(z, 1))$ then

$$T(p)G_k(z) = T(p)G_k(\Lambda(z, 1))$$

It's enough to show $G_k(\Lambda(z, 1))$ is an eigenvector.

$$\begin{aligned} T(p)G_k(\Lambda(z, 1)) &= p^{k-1}G_k(T(p)\Lambda) = p^{k-1}G_k\left(\sum_{[\Lambda, \Lambda'] = p} \Lambda'\right) \\ &= p^{k-1} \sum_{[\Lambda, \Lambda'] = p} G_k(\Lambda') = p^{k-1} \sum_{[\Lambda, \Lambda'] = p} \sum_{\substack{\lambda \in \Lambda' \\ \lambda \neq 0}} \frac{1}{\lambda^k} \end{aligned}$$

We know we have $(p+1)$ sublattices of index p , $[\Lambda : \Lambda'] = p \implies p\Lambda \subseteq \Lambda'$. If $\lambda \in p\Lambda$, then it appears $(p+1)$ times in the above double sum. If $\lambda \in \Lambda \setminus p\Lambda$, then there exists unique Λ' containing λ such that $[\Lambda, \Lambda'] = p$. ($\Lambda' = \lambda\mathbb{Z} + p\Lambda$). Suppose Λ', Λ'' are two such sublattices. Then $\lambda\mathbb{Z} + p\mathbb{Z} \subseteq \Lambda' \cap \Lambda''$. Then

$$p\Lambda \subsetneq \Lambda' \cap \Lambda'' \subseteq \Lambda', \Lambda'' \subsetneq \Lambda$$

$[\Lambda : p\Lambda] = p^2$ and $[\Lambda : \Lambda'] = p$ so $\Lambda' \cap \Lambda'' = \Lambda' = \Lambda''$. Then

$$\begin{aligned} p^{k-1} \sum_{[\Lambda, \Lambda'] = p} \sum_{\substack{\lambda \in \Lambda' \\ \lambda \neq 0}} \frac{1}{\lambda^k} &= p^{k-1} \left[(p+1) \sum_{\lambda \in p\Lambda} \frac{1}{\lambda^k} + \sum_{\lambda \in \Lambda \setminus p\Lambda} \frac{1}{\lambda^k} \right] \\ &= p^{k-1} \left[p \sum_{\lambda \in p\Lambda} \frac{1}{\lambda^k} + G_k(\Lambda) \right] \\ &= p^{k-1} \left[p \sum_{\lambda \in \Lambda} \frac{1}{(\lambda p)^k} + G_k(\Lambda) \right] \\ &= p^{k-1} \left[\frac{p}{p^k} G_k(\Lambda) + G_k(\Lambda) \right] \\ &= G_k(\Lambda) + p^{k-1} G_k(\Lambda) = \sigma_{k-1}(p) G_k(\Lambda). \end{aligned}$$

□

6 The Petersson Inner product

Definition 6.1. Let V be a finite dimensional $\mathbb{C}$ -vector space. A Hermitian inner product on V is a pairing:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$$

satisfying the following conditions:

- (i) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.
- (ii) $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$ for all $u, v \in V$ and for all $\lambda \in \mathbb{C}$.
- (iii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.
- (iv) $\langle u, u \rangle \geq 0$ for all $u \in V$ and $\langle u, u \rangle = 0 \Leftrightarrow u = 0$.

Lemma 6.2. Suppose Π is a commuting set of linear operators on V . Then:

- (1) The operators in Π have a common eigenvector.
- (2) If the operators in Π are self-adjoint with respect to the inner product on V , then V has an orthogonal basis consisting of common eigenvectors of all operators in Π .
- (3) If $T \in \Pi$ is self-adjoint with respect to the inner product on V , then all its eigenvalues are real.

Definition 6.3. Note that $F := f(z)\overline{g(z)}y^k$ satisfies $F(\gamma z) = F(z)$ for all $\gamma \in SL_2(\mathbb{Z})$. For $f, g \in S_k$, set

$$\langle f, g \rangle := \int_{\mathcal{F}} f(z)\overline{g(z)}y^k \frac{dx dy}{y^2}.$$

Lemma 6.4. $\frac{dx dy}{y^2}$ is GL_2^+ -invariant, i.e. given $U \subseteq \mathbb{H}$, $f : U \rightarrow \mathbb{C}$ a continuous function, and $\gamma \in GL_2^+(\mathbb{R})$,

$$\int_U f(z) \frac{dx dy}{y^2} = \int_{\gamma^{-1}U} f(\gamma z) \frac{dx dy}{y^2},$$

whenever the integral is well-defined.

Proof. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2^+(\mathbb{R})$. Then γ defines a holomorphic function on $\mathbb{H}$

$$g : \mathbb{H} \rightarrow \mathbb{H} \\ z \mapsto \frac{az + b}{cz + d},$$

since $cz + d \neq 0$ for all $z \in \mathbb{H}$. g can also be viewed as a real differentiable function on $\mathbb{H} \subseteq \mathbb{R}^2 : g(u(x, y), v(x, y))$, where $u(x, y) = \mathbb{R}(g(z))$, $v(x, y) = \mathbb{Im}(g(z))$. Hence,

$$\int_U f(z) \frac{dx dy}{y^2} = \int_{\gamma^{-1}U} f(\gamma z) |\det J(x, y)| \frac{dx dy}{(\mathbb{Im}(\gamma z))^2},$$

where

$$J(x, y) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

Since γ is holomorphic on $\mathbb{H}$, γ satisfies the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Thus

$$|\det J(x, y)| = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = |g'(z)|^2.$$

But $g'(z) = \frac{\det(\gamma)}{(cz+d)^2} = \frac{\mathbb{Im}(\gamma z)}{\gamma(z)}$. Putting everything together, we get

$$\int_{\gamma^{-1}U} f(\gamma z) |\det J(x, y)| \frac{dx dy}{(\mathbb{Im}(\gamma z))^2} = \int_{\gamma^{-1}U} f(\gamma z) \frac{dx dy}{y^2}$$

□

Recall from Problem sheet 1

$$(f|_k \gamma) = \det(\gamma)^{k/2} (cz + d)^{-k} f(\gamma z)$$

So for $\gamma \in GL_2^+(\mathbb{R})$, $f(\gamma z)\overline{g(\gamma z)}\mathbb{Im}(\gamma z)^k = (f|_k \gamma)(z)\overline{(g|_k \gamma)(z)}\mathbb{Im}(z)^k$. If $\gamma \in SL_2(\mathbb{Z})$, the the integral is $SL_2(\mathbb{Z})$ invariant. Final observation is that the integral converges if either f or g is a cusp form.

Say f is a cusp form. Then $|f| = |q|(a_1 + a_2 q + \dots)$ where $|q| = e^{-2\pi y} \rightarrow 0$ as $y \rightarrow \infty$. The integrand is continuous on $\mathcal{F}_y$ the fundamental domain under y hence bounded. The integrand is bounded on U_y , for y sufficiently large. We want to show the $T(n)$'s are self-adjoint with respect to $\langle \cdot, \cdot \rangle$. Enough to prove our goal to $T(p)$, p prime. Recall

$$(T(p)f)(z) = p^{k-1} \sum_{\eta \in S_p} d^{-k} f\left(\frac{az + b}{d}\right)$$

where $S_p = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \dots, \begin{pmatrix} 1 & p-1 \\ 0 & p \end{pmatrix}, \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \right\}$ and $\eta = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$. This is equivalent to

$$p^{k/2-1} \sum_{\mathcal{L} \in S_p} (f|_k \mathcal{L})(z)$$

But each term is not modular form in general, we now introduce the congruence subgroup.

Definition 6.5. Let $N \in \mathbb{Z}_{\geq 0}$, $\Gamma := SL_2(\mathbb{Z})$ then

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$\Gamma(N) < \Gamma$ is called the principal congruence subgroup of level N . $\Gamma(N)$ is the kernel of the homomorphism:

$$\text{red}_N : \Gamma \rightarrow SL_2(\mathbb{Z}/N\mathbb{Z})$$

$$\gamma \mapsto \gamma \pmod{N}$$

In addition, this map is indeed surjective. $\Gamma(N)$ is a normal subgroup of Γ i.e. $\gamma\Gamma(N)\gamma^{-1} \subseteq \Gamma(N)$ for all $\gamma \in \Gamma$. By the first isomorphism theorem,

$$\Gamma/\Gamma(N) \cong SL_2(\mathbb{Z}/N\mathbb{Z})$$

In particular, $[\Gamma : \Gamma(N)] = |SL_2(\mathbb{Z}/N\mathbb{Z})|$ which is finite.

Definition 6.6. A congruence subgroup is a subgroup Γ' of Γ such that $\Gamma(N) \subseteq \Gamma'$, for some $N \in \mathbb{Z}_{>0}$. In particular, $\Gamma(1) = SL_2(\mathbb{Z})$.

Example. Let

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma \mid N|c \right\}, \quad N \in \mathbb{Z}_{>0}$$

Then $\Gamma(N) \subseteq \Gamma_0(N) \subseteq \Gamma$.

Remark. $\Gamma(M) \supseteq \Gamma(N)$ if $M|N$ and $\Gamma_0(M) \supseteq \Gamma_0(N)$ if $M|N$.

Let Γ' be a congruence subgroup. Then $T^h \in \Gamma'$, for some h . Let f be a meromorphic function on $\mathbb{H}$ satisfying the modularity condition with weight k or Γ' . Since $T^h \in \Gamma'$ then $f(z+h) = f(z)$ for all $z \in \mathbb{H}$. Thus f has Laurent expansion of the form

$$f = \sum_{n \in \mathbb{Z}} a_n q_h^n$$

where $q_h = e^{2\pi iz/h}$. Given this, we can make sense of the notion of f holomorphic or meromorphic at ∞ . But this is not enough condition at ∞ . The idea is the following: The action of Γ on $\mathbb{H}$ comes from the fact that each $\gamma \in \Gamma$ defines an automorphism of $\hat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$.

$$\begin{aligned} \gamma : \hat{\mathbb{C}} &\rightarrow \hat{\mathbb{C}} \\ z &\mapsto \frac{az+b}{cz+d} \end{aligned}$$

where if $c \neq 0$, $\gamma\infty := \frac{a}{c}$ and $\gamma(-\frac{d}{c}) = \infty$. If $c = 0$ then $d \neq 0$, $\gamma\infty := \infty$. Consider $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$. Note if $z \in \mathbb{P}^1(\mathbb{Q})$ then $\gamma z \in \mathbb{P}^1(\mathbb{Q})$ for all $\gamma \in SL_2(\mathbb{Z})$. So we have an action $SL_2(\mathbb{Z})$ on $\mathbb{P}^1(\mathbb{Q})$. Under this action, $\infty \rightarrow \frac{a}{c} \in \mathbb{Q}$, if $c \neq 0$ then $\infty \rightarrow \infty$. Conversely, if $\frac{a}{c} \in \mathbb{Q}$ with $(a, c) = 1$ then there exists $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma\infty = \frac{a}{c}$ i.e. we can find b, d such that $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 1$ this is the γ we need. So all rational numbers are equivalent to ∞ under the action of $SL_2(\mathbb{Z})$ i.e.

$$SL_2(\mathbb{Z})_\infty = \mathbb{P}_1(\mathbb{Q})$$

But this is not the case for Γ' , a congruence subgroup. Identify points in $\mathbb{P}^1(\mathbb{Q})$ that are equivalent under the action of Γ' on $\mathbb{P}^1(\mathbb{Q})$ such that an equivalence class is called a cusp of Γ' . Notice that the set of cusps of Γ' is finite and

$$\Gamma' \backslash \mathbb{P}^1(\mathbb{Q}) \cong \Gamma' \backslash SL_2(\mathbb{Z}) / \text{Stab}_\infty(SL_2(\mathbb{Z}))$$

where $\text{Stab}_\infty(SL_2(\mathbb{Z})) = \langle \pm T \rangle$. In conclusion, the number of cusps of $\Gamma' \leq [SL_2(\mathbb{Z}) : \Gamma']$ hence finite.

Examples.

- (1) $\Gamma = SL_2(\mathbb{Z})$ has only one cusp at ∞ .
- (2) $\Gamma_0(p)$, $p \in \mathbb{Z}_{>0}$ prime then

$$\begin{aligned} \Gamma_0(p)\infty &= \left\{ \begin{pmatrix} a & b \\ pc & d \end{pmatrix} \infty \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma \right\} \\ &= \left\{ \frac{a}{pc} \mid (a, pc) = 1 \right\} \cup \{\infty\} \end{aligned}$$

$$\begin{aligned}\Gamma_0(p)0 &= \left\{ \begin{pmatrix} a & b \\ pc & d \end{pmatrix} 0 \mid \begin{pmatrix} a & b \\ pc & d \end{pmatrix} \in \Gamma \right\} \\ &= \left\{ \frac{b}{d} \mid (b, d) = 1, p \nmid d \right\}\end{aligned}$$

$\Gamma_0(p)$ has 2 cusps: $\infty, 0$. Take $p = 2$, notice that $\Gamma = \Gamma_0(2) \cup \Gamma_0(2)S \cup \Gamma_0(2)ST$. Then $[\Gamma : \Gamma_0(2)] = 3$. A fundamental domain for $\Gamma_0(z)$ is (up to potential overlaps on the translates of boundaries of $\mathcal{F}$) a set of "measure zero". If f satisfies the modularity condition with weight k for Γ' , a congruence subgroup then we say f is meromorphic or holomorphic at infinity if and only if f is meromorphic or holomorphic at all cusps of Γ' . Now all cusp are of the form $\gamma\infty$ with $\gamma \in SL_2(\mathbb{Z})$. Above is equivalent to $f(\gamma z)$ is meromorphic or holomorphic at ∞ for all $\gamma \in SL_2(\mathbb{Z})$ equivalent to

$$(cz + d)^k f|_k \gamma(z)$$

be meromorphic or holomorphic at ∞ . $(f|_k \gamma)(z)$ is meromorphic or holomorphic at ∞ . We can check this using Laurent expansions because $(f|_k \gamma)$ satisfies the modularity condition with weight k for $\gamma^{-1}\Gamma'\gamma$ which is another congruence subgroup

$$\Gamma(N) \subset \Gamma' \implies \Gamma(N) \subset \gamma^{-1}\Gamma'\gamma$$

Let $\gamma^{-1}\gamma'\gamma \in \gamma^{-1}\Gamma'\gamma$ then $(f|_k \gamma)|_k \gamma^{-1}\gamma'\gamma = f|_k(\gamma\gamma^{-1}\gamma'\gamma) = (f|_k \gamma')|_k \gamma = f|_k \gamma$.

Definition 6.7. A modular form of weight k for Γ' , a congruence subgroup is a function $f : \mathbb{H} \rightarrow \mathbb{C}$ satisfying

- (i) f is holomorphic on $\mathbb{H}$.
- (ii) $f|_k \gamma' = f$ for all $\gamma' \in \Gamma'$.
- (iii) $f|_k \gamma$ is holomorphic at ∞ for all $\gamma \in SL_2(\mathbb{Z})$.

If $f|_k \gamma$ vanishes at ∞ for all $\gamma \in SL_2(\mathbb{Z})$ then we call f a cusp form for Γ' .

Remark. If $\Gamma'' \subseteq \Gamma'$ then $M_k(\Gamma') \subseteq M_k(\Gamma'')$ and similarly for cusp form.

Here are some observations about corresponding fundamental domain. Let $\{\gamma_i\}_{i=0}^{[\Gamma':\Gamma'']}$ be a set of right coset representative of Γ'' in Γ' . Thus

$$\Gamma' = \bigcup_{i=1}^{[\Gamma':\Gamma'']} \Gamma'' \gamma_i$$

Then

$$\bigcup_{i=1}^{[\Gamma':\Gamma'']} \gamma_i \mathcal{F}(\Gamma')$$

is a fundamental domain for Γ'' where $\mathcal{F}(\Gamma')$ is a fundamental domain for Γ' .

Proof. Let $z \in \mathbb{H}$. We have to show there exists $\gamma'' \in \Gamma''$ such that $\gamma'' z \in \bigcup_i \gamma_i \mathcal{F}(\Gamma')$. We know there exists some $\gamma' \in \Gamma'$ such that $z = \gamma' z_0 \in \mathcal{F}(\Gamma')$. Since $\gamma' \in \Gamma'$ then $\gamma' = (\gamma'')^{-1} \gamma_i$ for some i and some $\gamma'' \in \Gamma''$. Thus,

$$\gamma'' z = (\gamma'')(\gamma'')^{-1} \gamma_i z_0 = \gamma_i z_0 \in \gamma_i \mathcal{F}(\Gamma')$$

Now suppose there exists $z, z' \in \bigcup_i \gamma_i \mathcal{F}(\Gamma')^\circ$ such that $z = \gamma'' z'$ for some $\gamma'' \in \Gamma''$. We want to show $\gamma'' = \pm I_2$. There exists i, j and $z_0, z'_0 \in \mathcal{F}(\Gamma')$ such that

$$z = \gamma_i z_0, \quad z' = \gamma_j z'_0$$

Then $\gamma_i z_0 = \gamma'' \gamma_j z'_0$ which means $z_0 = \gamma_i^{-1} \gamma'' \gamma_j z'_0$. But $\gamma_i^{-1} \gamma'' \gamma_j \in \Gamma'$ but $z_0, z'_0 \in \mathcal{F}(\Gamma')^\circ$ i.e. $\gamma_i^{-1} \gamma'' \gamma_j = \pm I_2$. $\gamma_i = \pm \gamma'' \gamma_j$ which means $\gamma'' = \pm I_2$ since they represent different cosets of Γ'' in Γ' . $\square$

Definition 6.8. Let $\Gamma' \subset \Gamma$ be a congruence subgroup. We define the Petersson inner product on $S_k(\Gamma')$ to be

$$\begin{aligned}\langle \cdot, \cdot \rangle_{\Gamma'} : S_k(\Gamma') \times S_k(\Gamma') &\rightarrow \mathbb{C} \\ f, g &\mapsto \frac{1}{[\Gamma : \Gamma']} \int_{\mathcal{F}(\Gamma')} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2}\end{aligned}$$

This converges if either f or g is a cusp form. First the choice of the fundamental domain $\mathcal{F}(\Gamma')$ doesn't matter similar to the case of $\Gamma = SL_2(\mathbb{Z})$. So let

$$\mathcal{F}(\Gamma') = \bigsqcup_i \gamma_i \mathcal{F}(\Gamma)$$

where $\Gamma = \bigcup_i \Gamma' \gamma_i$. Then

$$\begin{aligned}\int_{\mathcal{F}(\Gamma')} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2} &= \sum_i \int_{\gamma_i \mathcal{F}(\Gamma)} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2} \\ &= \sum_i \int_{\mathcal{F}(\Gamma)} (f|_k \gamma_i(z)) \overline{(g|_k \gamma_i(z))} y^k \frac{dx dy}{y^2}\end{aligned}$$

At least one of them is 0 at infinity, i.e. is $O(e^{-2\pi y/h})$ for some $h \in \mathbb{Z}_{>0}$. Thus, the integral converges.

Lemma 6.9. Let Γ', Γ'' be congruence subgroup such that $\Gamma'' \subset \Gamma'$. Let $f, g \in M_k(\Gamma')$ such that at least one of them is a cusp form. Then

$$\langle f, g \rangle_{\Gamma''} = \langle f, g \rangle_{\Gamma'}$$

Proof.

$$\begin{aligned} [\Gamma : \Gamma''] \langle f, g \rangle_{\Gamma''} &= \int_{\mathcal{F}(\Gamma'')} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2} \\ &= \sum_i \int_{\gamma_i \mathcal{F}(\Gamma')} f|_k \gamma_i(z) \overline{g|_k \gamma_i(z)} y^k \frac{dx dy}{y^2} \\ &= [\Gamma' : \Gamma''] [\Gamma : \Gamma'] \langle f, g \rangle_{\Gamma'} \end{aligned}$$

□

Lemma 6.10. Let $\alpha \in M_2(\mathbb{Z})$, $\det(\alpha) = D > 0$. Let $N > 0$ and $\Gamma' = \alpha\Gamma(DN)\alpha^{-1}$. Then

- (1) Γ' is a congruence subgroup. We have $\Gamma(D^2N) \subset \Gamma' \subset \Gamma(N)$. A fundamental domain for Γ' is $\alpha\mathcal{F}(\Gamma(DN))$ where $\mathcal{F}(\Gamma(DN))$ is a fundamental domain for $\Gamma(DN)$.
- (2) If $f \in M_k(\Gamma(N))$ then $f|_k \alpha \in M_k(\Gamma(DN))$. If $g \in M_k(\Gamma(N))$ then $g|_k \alpha^{-1} \in M_k(\Gamma')$. The above holds with M_k replaced by S_k . Furthermore,

$$\langle f, g \rangle_{\Gamma(N)} = \langle f|_k \alpha, g|_k \alpha^{-1} \rangle_{\Gamma(DN)}$$

Equivalently, $\langle f|_k \alpha, g \rangle_{\Gamma(DN)} = \langle f, g|_k \alpha^{-1} \rangle_{\Gamma'}$.

Proof.

- (1) Let $\gamma' \in \Gamma'$ then we know that $\det(\gamma') = 1$. Let $\tilde{\alpha} = D\alpha^{-1} \in M_2(\mathbb{Z})$ with $\det(\tilde{\alpha}) = D$. Let $\gamma \in \Gamma(DN)$, $\alpha\gamma\tilde{\alpha} \in M_2(\mathbb{Z})$. Note that $\alpha\gamma\tilde{\alpha} \equiv \alpha\tilde{\alpha} \pmod{DN} \equiv DI_2 \pmod{DN}$. Thus,

$$\alpha\gamma \frac{1}{D} \tilde{\alpha} \equiv I_2 \pmod{N}$$

i.e. $\alpha\gamma\alpha^{-1} \equiv I_2 \pmod{N}$. Therefore, $\alpha\gamma\alpha^{-1} \in \Gamma(N)$. We have $\alpha\Gamma(DN)\alpha^{-1} \subset \Gamma(N)$ for any matrix α with $\det(\alpha) = D$. Replace N with DN then

$$\alpha\Gamma(D^2N)\alpha^{-1} \subset \Gamma(DN)$$

$$\tilde{\alpha}\Gamma(D^2N)\tilde{\alpha}^{-1} \subset \Gamma(DN) \implies \Gamma(D^2N) \subset (\tilde{\alpha})^{-1}\Gamma(DN)\tilde{\alpha} = \Gamma'$$

The fundamental domain part: want to show $\alpha\mathcal{F}(\Gamma(DN))$ is a fundamental domain for Γ' . Let $z \in \mathbb{H}$, we have to show there exists $\gamma' = \alpha\gamma\alpha^{-1} \in \Gamma'$ with $\gamma \in \Gamma(DN)$ such that $\gamma'z \in \alpha\mathcal{F}(\Gamma(DN))$. We know there exists $\gamma \in \Gamma(DN)$ such that $\gamma(\alpha^{-1}z) \in \mathcal{F}(\Gamma(DN))$, $\alpha\gamma(\alpha^{-1}z) \in \alpha\mathcal{F}(\Gamma(DN))$. Now suppose there exists $z, z' \in \mathcal{F}(\Gamma(DN))^\circ$ such that $z = \gamma'z'$ for some $\gamma' = \alpha\gamma\alpha^{-1} \in \Gamma'$ which implies $\alpha^{-1}z = \gamma\alpha^{-1}z'$ which implies $\gamma = \pm I_2$ and $\gamma' = \pm I_2$.

- (2) Let $f \in M_k(\Gamma(N))$ and $\gamma \in \Gamma(DN)$. We want to show that $(f|_k \alpha)|_k \gamma = f|_k \alpha$. Note that

$$(f|_k \alpha)|_k \gamma = f|_k \alpha\gamma = f|_k (\gamma'\alpha) = (f|_k \gamma')|_k \alpha = f|_k \alpha$$

since $\alpha\gamma\alpha^{-1} \in \Gamma' \subset \Gamma(N) \implies \alpha\gamma = \gamma'\alpha$ with $\gamma' \in \Gamma' \subset \Gamma(N)$.

Now let $g \in M(\Gamma(N))$, $\gamma' = \alpha\gamma\alpha^{-1} \in \Gamma'$, $\gamma \in \Gamma(DN)$.

$$(g|_k \alpha^{-1})|_k (\alpha\gamma\alpha^{-1}) = (g|_k \gamma)|_k \alpha^{-1} = g|_k \alpha^{-1}.$$

We also have to check the behaviour at the cusps. e.g. we need to check that $(f|_k \alpha)|_k \gamma$ is holomorphic at ∞ , for all $\gamma \in SL_2(\mathbb{Z})$. The idea is that $\alpha\gamma = AU$ where $A \in SL_2(\mathbb{Z})$ and $U \in GL_2^+(\mathbb{Q})$ upper triangular. Let $F = f|_k A$, F is holomorphic at ∞ since $f \in M_k(\Gamma(N))$. Thus, F has the q -expansion

$$F = \sum_{m \geq 0} c(m) e^{2\pi i z m / h}, \quad h \in \mathbb{Z}_{>0}$$

Then

$$F|_k U = \sum_{m \geq 0} c(m) e^{2\pi i (az+b)/dh}$$

which is holomorphic at ∞ as well.

$$\begin{aligned} [\Gamma : \Gamma(DN)] \langle f|_k \alpha, g|_k \alpha^{-1} \rangle_{\Gamma(DN)} &= \int_{\mathcal{F}(\Gamma(DN))} (f|_k \alpha)(z) \overline{g|_k \alpha^{-1}(z)} y^k \frac{dx dy}{y^2} \\ &= \int_{\alpha\mathcal{F}(\Gamma(DN))} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2} \\ &= [\Gamma : \Gamma'] \langle f, g \rangle_{\Gamma'} \\ &= [\Gamma : \Gamma(DN)] \langle f, g \rangle_{\Gamma(N)} \end{aligned}$$

for the second part, replace f with $f|_k \alpha$ and α with $\tilde{\alpha} = D\alpha^{-1}$ in the proof above. Also note that $g|_k \alpha^{-1} = g|_k \alpha$ because $g|_k \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix} = g$.

□

Theorem 6.11. The Hecke operators are self-adjoint with respect to $\langle \cdot, \cdot \rangle$ on $S_k(SL_2(\mathbb{Z}))$. So $S_k(SL_2(\mathbb{Z}))$ has an orthogonal basis of eigenforms. Also the eigenvalues of the Hecke operators $T(n)$, $n \in \mathbb{Z}_{>0}$ are algebraic integers.

Proof. Enough to show self-adjointness for $T(p)$, $p \in \mathbb{Z}_{>0}$ prime. Recall

$$T(p)f = p^{k/2-1} \sum_{\gamma \in S_p} f|_k \gamma$$

Note $\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. So every $\gamma \in S_p$ is of the form $u\alpha v$ with $u, v \in SL_2(\mathbb{Z})$ and $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. Then let $f, g \in M_k$, at least one a cusp form

$$\begin{aligned} \langle f|_k u\alpha v, g \rangle_{\Gamma(p)} &= \langle f|_k \alpha v, g \rangle_{\Gamma(p)} \\ &= \langle f, g|_k (\gamma v)^{-1} \rangle_{\Gamma(p^2)} \\ &= \langle f, g|_k \alpha^{-1} \rangle_{\Gamma(p^2)} \\ &= \langle f|_k \alpha, g \rangle_{\Gamma(p)} \end{aligned}$$

Similarly, we have

$$\langle f, g|_k u\alpha v \rangle_{\Gamma(p)} = \langle f, g|_k \alpha \rangle_{\Gamma(p)}$$

Then

$$\begin{aligned} \langle T(p)f, g \rangle &= p^{k/2-1}(p+1) \langle f|_k \alpha, g \rangle \\ &= p^{k/2-1}(p+1) \langle f, g|_k \alpha^{-1} \rangle \\ &= p^{k/2-1}(p+1) \langle f, g|_k \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \rangle \\ &= \langle f, T(p)g \rangle \end{aligned}$$

□

7 L -functions

Let $\Gamma = SL_2(\mathbb{Z})$, $M_k(\Gamma) =: M_k$.

Definition 7.1. Let $f = \sum_{n \geq 1} a_n q^n \in M_k$ then the L -function associated to f is $L(f, s)$ is $\sum_{n \geq 1} a_n n^{-s}$, $s \in \mathbb{C}$, $\text{Re}(s) \gg 0$.

Theorem 7.2. If $f \in M_k$ then $L(f, s)$ converges absolutely and uniformly on compact subsets of $\{s \in \mathbb{C} \mid \text{Re}(s) > k\}$ and hence $L(f, s)$ is analytic function for $\text{Re}(s) > k$. If $f \in S_k$ then the above holds for $\text{Re}(s) > k/2 + 1$.

Proof. If $f = \sum_{n \geq 0} a_n q^n \in M_k \setminus S_k$ then

$$|a_n| \leq cn^{k-1} \text{ for some } c \in \mathbb{R}_{>0} \text{ and } n \text{ large enough}$$

then $|a_n n^{-s}| = |a_n| n^{-\text{Re}(s)}$. Let $\text{Re}(s) = k + \varepsilon$, $\varepsilon > 0$ then

$$|a_n| n^{-\text{Re}(s)} \leq cn^{k-1} n^{-k-\varepsilon} = c^{-(\varepsilon+1)}$$

for n large enough. Note that $\sum_{n \geq 1} n^{-(\varepsilon+1)}$ converges so $L(f, s)$ converges absolutely and uniformly on compact subsets $\alpha \leq \text{Re}(s) \leq \beta$, $\alpha, \beta \in \mathbb{R}_{>0}$.

For $f \in S_k$, we have $|a_n| = O(n^{k/2})$. In fact, by Deligne's result ($|a_n| = O(n^{k-1/2 + \varepsilon})$, $\forall \varepsilon > 0$) we can do better and get that $L(f, s)$ is analytic for $\text{Re}(s) > \frac{k+1}{2}$. $\square$

Theorem 7.3. If $f = \sum_{n \geq 0} a_n q^n \in M_k$ is a normalized eigenform then $L(f, s)$ has an Euler product expansions of the form

$$L(f, s) = \prod_{\substack{p \in \mathbb{Z}_{>0} \\ p \text{ prime}}} \frac{1}{(1 - a_p p^{-s} + p^{k-1-2s})}, \quad \forall s$$

in the region of absolute convergence. Conversely, if $f \in M_k$ and $L(f, s)$ has an Euler product expansions as above then f is a normalized eigenform.

Proof. ($\implies$): If f is a normalized ($a_1 = 1$) eigenform then $a_m a_n = a_{mn}$ whenever $(m, n) = 1$. Then

$$\begin{aligned} L(f, s) &= \sum_{n \geq 1} a_n n^{-s} \\ &= \prod_{\substack{p \in \mathbb{Z}_{>0} \\ p \text{ prime}}} \sum_{m \geq 0} a_p^m p^{-ms} \end{aligned}$$

We need to show the product converges and

$$\sum_{m \geq 0} a_p^m p^{-ms} = \frac{1}{(1 - a_p p^{-s} + p^{k-1-2s})}$$

Equivalently,

$$\begin{aligned} (1 - a_p p^{-s} + p^{k-1-2s})(a_1 + a_p p^{-s} + a_{p^2} p^{-2s} \dots) &= 1 \\ 1 + p^{-s}(1 - 1) + p^{-2s}(a_{p^2} - a_p a_p + p^{k-1}) + \dots &= 1 \end{aligned}$$

but $a_{p^2} - a_p a_p + p^{k-1} = 0$ since f is an normalized eigenform. In general,

$$1 + \sum_{m \geq 2} \underbrace{p^{-ms}(a_{p^m} - a_p a_{p^{m-1}} + p^{k-1} a_{p^{m-2}})}_0$$

Now we show that the product is indeed convergent. Let $N \in \mathbb{Z}_{>0}$ and

$$\prod_{\substack{p \in \mathbb{Z}_{>0} \\ p \leq N}} \sum_{\substack{\text{prime } m \geq 0}} a_p^m p^{-ms} = \sum_{n \leq N} a_n n^{-s} + \underbrace{\sum_{\substack{n > N \\ n \text{ factorise into primes } p < N}} a_n n^{-s}}_{R(N, s)}$$

We want to show $R(N, s) \rightarrow 0$ as $N \rightarrow +\infty$. Notice that

$$|R(N, s)| \leq \sum_{\substack{n > N \\ n \text{ factorise into primes } p < N}} |a_n n^{-s}| \leq \sum_{n > N} |a_n n^{-s}|$$

But we see that $L(f, s)$ converges absolutely, thus $\sum_{n \geq 1} |a_n n^{-s}|$ converges. Then clearly $\sum_{n > N} |a_n n^{-s}| \rightarrow 0$ as $N \rightarrow \infty$ and we are done. $\square$

Examples.

(1)

$$L(\Delta, s) = \prod_p \frac{1}{1 - \tau(p)p^{-s} - p^{11-2s}}$$

(2)

$$L(\mathbb{G}_k, s) = \prod_k \frac{1}{1 - \sigma_{k-1}(p)p^{-s} + p^{k-1-2s}} = \prod_p \frac{1}{(1 - p^{-s})(1 - p^{k-1-s})} = \zeta(s)\zeta(s - k + 1)$$

We heard (and we'll see next week) that $\zeta(s)$ has an analytic continuation to $\mathbb{C}$ except for a simple pole at $s = 1$, with residue equals 1. Thus, $L(\mathbb{G}_k, s)$ has analytic continuation to $\mathbb{C}$ except for possible poles at $s = 1$ and $s = k$. For $s = 1$, since $\zeta(2 - k) = 0$ we do not get a pole at $s = 1$. For $s = k$, $\zeta(k) \neq 0$ so we do get a simple pole at $s = k$.

Definition 7.4. For $s \in \mathbb{C}, \operatorname{Re}(s) > 0$ the gamma function is defined by

$$\Gamma(s) := \int_0^{+\infty} e^{-t} t^s \frac{dt}{t}$$

the integral converges absolutely and uniformly on compact subsets of $\{s \in \mathbb{C} \mid \operatorname{Re}(s) > 0\}$ and so $\Gamma(s)$ is analytic for $\operatorname{Re}(s) > 0$. In particular, $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(1) = 1$, $\Gamma(n+1) = n!$ for all $n \in \mathbb{Z}_{\geq 0}$. We can use $\Gamma(s+1) = s\Gamma(s)$ to get an analytic continuation of Γ to $\mathbb{C}$.

For $\operatorname{Re}(s) > -1$, let $\Gamma_1(s) := \frac{\Gamma(s+1)}{s}$ analytic except for $s = 0$ we get a simple pole. And by definition, $\Gamma_1(s) = \Gamma(s)$ for all $\operatorname{Re}(s) > 0$. Keep iterating, $\Gamma_2(s) := \frac{\Gamma_1(s+1)}{s+1} = \frac{\Gamma(s+2)}{s(s+1)}$ which is analytic for $\operatorname{Re}(s) > -2$ except simple poles at $s = 0$. Then

$$\Gamma_k(s) = \frac{\Gamma(s+k)}{s(s+1)\cdots(s+k-1)}$$

analytic for $\operatorname{Re}(s) > -k$ except for simple poles at $s = -n, n \in \mathbb{Z}_{\geq 0}$. $\Gamma_k(s) = \Gamma(s)$ for $\operatorname{Re}(s) > k$. The Γ_k 's gives us a meromorphic continuation of Γ to $\mathbb{C}$, analytic everywhere except for simple poles at $-n, n \in \mathbb{Z}_{\geq 0}$ with residues $\frac{(-1)^n}{n!}$.

Definition 7.5 (The Mellin Transform). Let $f : \mathbb{R}_{>0} \rightarrow \mathbb{C}$ continuous. The Mellin transform of f is

$$\mathcal{M}(f, s) := \int_0^{+\infty} f(t) t^s \frac{dt}{t}$$

whenever the integral converges. In particular, for $f(t) = e^{-t}$ then $\mathcal{M}(e^{-t}, s) = \Gamma(s)$.

Lemma 7.6. With the notation in the definition above, suppose the followings hold:

- (i) $f(t)t^N \rightarrow 0$ for all $N \in \mathbb{Z}_{>0}$ as $t \rightarrow \infty$.
- (ii) $\exists m \in \mathbb{R}_{>0}$ such that $|t^m f(t)|$ is bounded as $t \rightarrow 0$.

Then $\mathcal{M}(f, s)$ is analytic for $\operatorname{Re}(s) > m$.

Definition 7.7. For $f \in S_k$, the Mellin Transform of f is given by

$$\mathcal{M}(f, s) := \mathcal{M}(f(iy), s) = \int_0^{+\infty} f(iy) y^s \frac{dy}{y}$$

Theorem 7.8. Let $f \in S_k$, $f = \sum_{n \geq 1} a_n q^n$. Then

- (1) $\mathcal{M}(f, s)$ is an analytic function on $\mathbb{C}$.
- (2) $\mathcal{M}(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$. Hence $L(f, s)$ has analytic continuation to $\mathbb{C}$. Furthermore, $L(f, -n) = 0$ for all $n \in \mathbb{Z}_{\geq 0}$.
- (3) $\mathcal{M}(f, k-s) = (-1)^{k/2} \mathcal{M}(f, s)$ is the functional equation.

Proof.

- (1) $f(iy)$ is $O(e^{-2\pi y})$ so it's analytic at infinity. At 0, using the transformation of f with respect to $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ then

$$f\left(\frac{-1}{iy}\right) = f\left(\frac{i}{y}\right) = (iy)^k f(iy)$$

Now use the above, noting that $y \rightarrow \infty, \frac{1}{y} \rightarrow 0$. Thus, f is analytic at 0. Thus, $\mathcal{M}(f, s)$ is an analytic function on $\mathbb{C}$.

(2) By computation

$$\begin{aligned}
\mathcal{M}(f, s) &= \int_0^\infty \sum_{n \geq 1} a_n e^{-2\pi n y} y^s \frac{dy}{y} \\
&= \sum_{n \geq 1} a_n \int_0^\infty e^{-2\pi n y} y^s \frac{dy}{y} \\
&= \sum_{n \geq 1} a_n \int_0^\infty e^{-y} (2\pi n)^{-s} y^s \frac{dy}{y} \\
&= \sum_{n \geq 1} a_n n^{-s} (2\pi)^{-s} \Gamma(s) \\
&= (2\pi)^{-s} \Gamma(s) L(f, s).
\end{aligned}$$

(3) By computation

$$\begin{aligned}
\mathcal{M}(f, k-s) &= \int_0^\infty f(iy) y^{s-k} \frac{dy}{y} \\
&= \int_\infty^0 f\left(\frac{i}{y}\right) y^{s-k} \left(-\frac{dy}{y}\right) \\
&= \int_0^\infty (iy)^k f(iy) y^{s-k} \frac{dy}{y} \\
&= (-1)^{k/2} \mathcal{M}(f, s)
\end{aligned}$$

□

Exercise. For $\operatorname{Re}(s) > 0$,

$$\mathcal{M}\left(\frac{t}{e^t - 1}, s\right) = \Gamma(s+1) \zeta(s+1).$$

Questions:

- (1) Can we show analytic continuation of $\zeta(z)$ using similar methods.
- (2) Do we have a converse i.e. given a Dirichlet L -series $\sum_{n \geq 1} a_n n^{-s}$, can we tell when it comes from a modular form for $SL_2(\mathbb{Z})$.

To answer these questions we need to think more about the Mellin transform and the relation to the Fourier Transform.

Definition 7.9. Let $f \in L^1(\mathbb{R})$ then we define the Fourier transform of f as

$$\hat{f}(x) = \mathcal{F}[(f)](x) = \int_{\mathbb{R}} f(t) e^{-2\pi i t x} dt$$

Theorem 7.10 (Fourier Inversion). If $f, \hat{f} \in L^1(\mathbb{R})$ then for all $x \in \mathbb{R}$,

$$f(x) = \int_{\mathbb{R}} \hat{f}(t) e^{2\pi i t x} dt = \hat{\hat{f}}(-x)$$

Example (Schwarz Space). $\mathcal{S}(\mathbb{R}) = \{f \in C^\infty(\mathbb{R}) \mid x^N f^{(n)}(x) \rightarrow 0, \forall N, n \in \mathbb{Z}_{\geq 0}\}$ as $x \rightarrow \infty$.

Proposition 7.11. Properties:

- (1) If $f, g \in L^1(\mathbb{R})$, $a, b \in \mathbb{R}$:

$$\widehat{af + bg} = a\hat{f} + b\hat{g}$$

- (2) If $f \in L^1(\mathbb{R})$, $g(t) = f(at)$ for some $a \in \mathbb{R}_{>0}$ then

$$\hat{g}(x) = \frac{1}{a} \hat{f}\left(\frac{x}{a}\right)$$

- (3) If $f \in L^1(\mathbb{R})$ and $g(t) = -2\pi i f(t) \in L^1(\mathbb{R})$ then

$$\hat{g}(x) = \frac{d}{dx}(\hat{f}(x))$$

- (4) If f is continuous differentiable on $\mathbb{R}$ and $f, f' \in L^1(\mathbb{R})$ then

$$\hat{f}'(x) = 2\pi i x \hat{f}(x)$$

Example. $f(x) = e^{-\pi x^2} \in \mathcal{S}(\mathbb{R})$ and $\hat{f}(x) = e^{-\pi x^2}$.

Theorem 7.12 (Poisson Summation). Suppose f is continuous on $\mathbb{R}$ and $f \in L^1(\mathbb{R})$ then

$$\sum_{n \in \mathbb{Z}} f(n+x) = \sum_{m \in \mathbb{Z}} \hat{f}(m) e^{2\pi i m x}$$

provided both series converge absolutely and uniformly. In particular for $x = 0$ we have

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{m \in \mathbb{Z}} \hat{f}(m)$$

Proof. Let $g(x) := \sum_{n \in \mathbb{Z}} f(n+x)$ then g is continuous on $\mathbb{R}$ and $g(x+1) = g(x)$ for all $x \in \mathbb{R}$. g has Fourier series expansion

$$g(x) \sim \sum_{m \in \mathbb{Z}} c_m e^{2\pi i m x}, \text{ where } c_m = \int_0^1 g(x) e^{-2\pi i m x} dx$$

Then

$$\begin{aligned} c_m &= \int_0^1 \sum_{n \in \mathbb{Z}} f(x+n) e^{-2\pi i m x} dx \\ &= \sum_{n \in \mathbb{Z}} \int_0^1 f(x+n) e^{-2\pi i m x} dx \\ &= \sum_{n \in \mathbb{Z}} \int_n^{n+1} f(t) e^{-2\pi i m t} dt \\ &= \int_{\mathbb{R}} f(t) e^{2\pi i m t} dt = \hat{f}(m) \end{aligned}$$

Since $\sum_{m \in \mathbb{Z}} |c_m| = \sum_{m \in \mathbb{Z}} |\hat{f}(m)|$, thus $g(x) = \sum_{m \in \mathbb{Z}} \hat{f}(m) e^{2\pi i m x}$. □

Example. For $t \in \mathbb{R}_{>0}$, let

$$\theta(t) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t} = 1 + 2 \sum_{n \in \mathbb{Z}_{>0}} e^{-\pi n^2 t}$$

Lemma 7.13. $\theta(\frac{1}{t}) = \sqrt{t} \theta(t)$ for all $t \in \mathbb{R}_{>0}$.

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{-\pi x^2 t}$, note that

$$\theta(t) = \sum_{n \in \mathbb{Z}} f(n)$$

We want to apply the Poisson summation. Let $g(x) = e^{-\pi x^2}$ then $\hat{g}(x) = e^{-\pi x^2}$ so

$$\hat{f}(x) = \frac{1}{\sqrt{t}} \hat{g}\left(\frac{x}{\sqrt{t}}\right) = \frac{1}{\sqrt{t}} e^{-\pi x^2/t}$$

Then

$$\sum_{m \in \mathbb{Z}} \hat{g}(m) = \frac{1}{\sqrt{t}} \sum_{m \in \mathbb{Z}} e^{-\pi m^2/t} = \frac{1}{\sqrt{t}} \theta\left(\frac{1}{t}\right)$$

By Poisson summation, we get $\theta(t) = \frac{1}{\sqrt{t}} \theta\left(\frac{1}{t}\right)$. □

We cannot take the Mellin transform of

$$\theta(t) = 1 + 2 \sum_{n \geq 1} e^{-\pi n^2 t} = O(e^{-\pi t})$$

Take instead $\omega(t) := \frac{\theta(t)-1}{2}$ then

$$\omega\left(\frac{1}{t}\right) = \frac{\theta(1/t)-1}{2} = \frac{\sqrt{t}\theta(t)-1}{2} = \frac{\sqrt{t}(\theta(t)-1)}{2} + \frac{\sqrt{t}}{2} - \frac{1}{2} = \sqrt{t}\omega(t) + \frac{\sqrt{t}}{2} - \frac{1}{2}$$

At $t = 0$, $\omega(t) = \frac{1}{\sqrt{t}} \omega\left(\frac{1}{t}\right) - \frac{1}{2} + \frac{1}{2\sqrt{t}}$

$$\implies \sqrt{t}\omega(t) = \omega\left(\frac{1}{t}\right) - \frac{\sqrt{t}}{2} + \frac{1}{2}$$

as $t \rightarrow 0$, RHS is bounded. Thus, $\mathcal{M}(\omega, s)$ is analytic function for $\text{Re}(s) > 1/2$. For $\text{Re}(s) > 1$ then

$$\begin{aligned} \mathcal{M}\left(\omega, \frac{s}{2}\right) &= \int_0^{+\infty} \omega(t) t^{s/2} \frac{dt}{t} = \int_0^\infty \sum_{n \geq 1} e^{-\pi n^2 t} t^{s/2} \frac{dt}{t} \\ &= \sum_{n \geq 1} \int_0^\infty e^{-\pi n^2 t} t^{s/2} \frac{dt}{t} = \sum_{n \geq 1} \int_0^\infty e^{-y} \left(\frac{y}{\pi n^2}\right)^{s/2} \frac{dy}{y} \\ &= \pi^{-s/2} \sum_{n \geq 1} n^{-s} \Gamma\left(\frac{s}{2}\right) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) \end{aligned}$$

for $\operatorname{Re}(s) > 1$. We claim that $\mathcal{M}(\omega, s/2)$ has analytic continuation to $\mathbb{C}$ except for simple poles at $s = 0$ and $s = 1$. Furthermore

$$\mathcal{M}(\omega, \frac{s}{2}) = \mathcal{M}(\omega, \frac{1-s}{2})$$

The fundamental equation for the Riemann ζ -function.

Proof. The trick:

$$\mathcal{M}(\omega, \frac{s}{2}) = \int_0^1 \omega(t) t^{s/2} \frac{dt}{t} + \int_1^\infty \omega(t) t^{s/2} \frac{dt}{t}$$

For $\operatorname{Re}(s) > 1$ we have

$$\begin{aligned} \int_0^1 \omega(t) t^{s/2} \frac{dt}{t} &= \int_\infty^1 \omega(\frac{1}{t}) t^{-s/2} \frac{-dt}{t} \\ &= \int_1^\infty \sqrt{t} \omega(t) t^{-s/2} \frac{dt}{t} + \int_1^\infty \sqrt{t} (\frac{\sqrt{t}}{2} - \frac{1}{2}) t^{-s/2} \frac{dt}{t} \\ &= \int_1^\infty \sqrt{t} \omega(t) t^{-s/2} \frac{dt}{t} + \frac{1}{s-1} - \frac{1}{s} \end{aligned}$$

For $\operatorname{Re}(s) > 1$, $\mathcal{M}(\omega, \frac{s}{2}) = \int_1^\infty \omega(t) (t^{(1-s)/2} + t^{s/2}) \frac{dt}{t} + \frac{1}{s-1} - \frac{1}{s}$ on $s \in \mathbb{C}$ so a simple pole at 1 with residue 1 and simple pole at 0 with residue 0. The RHS gives us the analytic continuation to $\mathbb{C}$ of $\mathcal{M}(\omega, s/2)$ except for the simple poles at $x = 0$ and $x = 1$. Also $\mathcal{M}(\omega, s/2) = \mathcal{M}(\omega, (1-s)/2)$. Thus, $\zeta(s)$ has analytic continuation except for the possible poles at $s = 1$ and $s = 0$. At $s = 0$, $\Gamma(s)$ has a simple pole at $s = 1$ with residue 1 so $\Gamma(s/2)$ has residue 2. Thus, $\zeta(0) = -\frac{1}{2}$. At $s = 1$, $\Gamma(1/2) = \sqrt{\pi}$ so Riemann zeta function has a simple pole at $s = 1$ with residue = 1. $\square$

Theorem 7.14 (Mellin Inversion). Suppose $\phi : \mathbb{R}_{>0} \rightarrow \mathbb{C}$ is twice continuously differentiable and suppose there exists some $\sigma \in \mathbb{R}$ such that $y^\sigma \phi(y), y^{\sigma+1} \phi'(y), y^{\sigma+2} \phi''(y) \in L^1(\mathbb{R}_{>0})$. Then

$$\mathcal{M}(\phi, s) := \int_0^{+\infty} \phi(y) y^s \frac{dy}{y}$$

converges for $\operatorname{Re}(s) = \sigma$ and $|\mathcal{M}(\phi, \sigma + ix)|$ is bounded and $\sim |x|^{-2}$ as $x \rightarrow \pm\infty$. Furthermore,

$$\phi(y) = \frac{1}{2\pi i} \int_{\operatorname{Re}(s)=\sigma} \mathcal{M}(\phi, s) y^{-s} ds$$

Proof. A change of variables and Fourier Inversion. Note that if $y \in \mathbb{R}_{>0}$, there exists unique $u \in \mathbb{R}$ such that $y = e^{2\pi u}$. Let $g(u) = e^{2\pi\sigma u} \phi(e^{2\pi u}) = y^\sigma \phi(y)$. Then g is twice continuously differentiable on $\mathbb{R}$ and $g, g', g'' \in L^1(\mathbb{R})$. Then

$$\begin{aligned} \hat{g}(-x) &= \int_{-\infty}^\infty g(u) e^{-2\pi i(-x)u} du \\ &= \int_{-\infty}^\infty y^\sigma \phi(y) y^{ix} du, \text{ where } y = e^{2\pi u} \\ &= \frac{1}{2\pi} \int_0^\infty \phi(y) y^{\sigma+ix} \frac{dy}{y} \\ &= \frac{1}{2\pi} \mathcal{M}(\phi, \sigma + ix) \end{aligned}$$

Also $\hat{g}''(-x) = (2\pi ix)^2 \hat{g}(-x)$, thus

as $x \rightarrow \pm\infty$. Then $\phi(y) = y^{-\sigma} g(u)$ where $u = \frac{\log(y)}{2\pi}$. Fourier Inversion gives $\hat{g}(-x) = g(x)$ which means

$$\begin{aligned} \phi(y) &= y^{-\sigma} \int_{-\infty}^\infty \frac{1}{2\pi} \mathcal{M}(\phi, \sigma + ix) e^{-2\pi i x u} dx \\ &= \int_{-\infty}^\infty \frac{1}{2\pi} \mathcal{M}(\phi, \sigma + ix) y^{-(\sigma+ix)} dx \\ &= \frac{1}{2\pi i} \int_{\operatorname{Re}(s)=\sigma} \mathcal{M}(\phi, s) y^{-s} ds \end{aligned}$$

$\square$

Theorem 7.15 (Hecke's Converse). Let $L(s) := \sum_{n \geq 1} a_n n^{-s}$ such that $a_n = O(n^A)$ for some $A \in \mathbb{R}_{>0}$. Let $\mathcal{M}(s) := (2\pi)^{-s} \Gamma(s) L(s)$. Suppose $\mathcal{M}(s)$ extends to a holomorphic function on $\mathbb{C}$ which is bounded on vertical strips (i.e. $\{s \in \mathbb{C} \mid \sigma_1 \leq \operatorname{Re}(s) \leq \sigma_2\}$, for all $\sigma_1, \sigma_2 \in \mathbb{R}, \sigma_1 < \sigma_2$) and $\mathcal{M}(s) = (-1)^{k/2} \mathcal{M}(k-s)$ for some $k \in \mathbb{Z}_{>2}$ even. Then $f = \sum_{n \geq 1} a_n q^n \in S_k(SL_2(\mathbb{Z}))$.

Proof. $a_n = O(n^A) \implies f$ is holomorphic on $\mathbb{H}$ since $f = \sum_{n \geq 1} a_n q^n$ converges absolutely and uniformly on compact

subsets of $\mathbb{H}$:

$$\sum_{n \geq 1} |a_n| |q^n| \leq c \sum_{n \geq 1} n^A e^{-2\pi n y} \leq c \int_0^\infty t^A e^{-2\pi y t} dt = c \int_0^\infty \left(\frac{t}{2\pi y}\right)^A e^{-t} \frac{dt}{2\pi y} = \frac{c}{(2\pi y)^{A+1}} \Gamma(A+1)$$

By definition, $f(z+1) = f(z)$ for all $z \in \mathbb{H}$. It remains to show $f(-1/z) = z^k f(z)$. It is enough to show this transformation for $z = iy$, $y \in \mathbb{R}_{>0}$. Let $g(z) := f(-1/z) - z^k f(z)$ then $g(z)$ is a holomorphic function on $\mathbb{H}$. If $g(iy) = 0$ for all $y \in \mathbb{R}_{>0}$ then $g(z) = 0$ for all $z \in \mathbb{H}$ since g is holomorphic and its set of zeros cannot have an accumulation point unless $g = 0$. We will show the transformation using the functional equation of $\mathcal{M}(s)$ and Mellin Inversion applied to $\mathcal{M}(s)$. For $\text{Re}(s) > 1 + A$, notice that $\mathcal{M}(s) = \mathcal{M}(f, s)$. $L(s) = \sum_{n \geq 1} a_n n^{-s}$ is holomorphic for $\text{Re}(s) > 1 + A$ i.e. it converges uniformly and absolutely on compact subsets of $\{s \in \mathbb{C} \mid \text{Re}(s) > 1 + A\}$. Let $\text{Re}(s) = 1 + A + \varepsilon$ then

$$\sum_{n \geq 1} |a_n| |n^{-s}| = \sum_{n \geq 1} |a_n| n^{-\text{Re}(s)} \leq c \sum_{n \geq 1} n^A n^{-(1+A+\varepsilon)} = c \sum_{n \geq 1} n^{-(1+\varepsilon)}$$

which converges. Then

$$\begin{aligned} \mathcal{M}(f, s) &= \int_0^\infty f(iy) y^s \frac{dy}{y} = \int_0^\infty \sum_{n \geq 1} a_n e^{-2\pi n y} y^s \frac{dy}{y} \\ &= \sum_{n \geq 1} a_n \int_0^\infty e^{-2\pi n y} y^s \frac{dy}{y} = (2\pi)^{-s} L(f, s) \Gamma(s) \end{aligned}$$

To apply the Mellin Inversion $\mathcal{M}(s)$, we need to check the conditions on previous Theorem is satisfied. Here $\phi(y) := f(iy) = \sum_{n \geq 1} a_n e^{-2\pi n y}$. Need to check $y^\sigma \phi(y), y^{\sigma+1} \phi'(y), y^{\sigma+2} \phi''(y)$ are $L^1(\mathbb{R}_{>0})$ for some $\sigma \in \mathbb{R}$. Notice that $\phi = O(e^{-2\pi y})$ as $y \rightarrow \infty$. By looking at $\sum_{n \geq 1} e^{-2\pi n y} = \frac{1}{1-e^{-2\pi y}}$. We shall consider about the Bernoulli series

$$\frac{x}{e^x - 1} = \sum_{n \geq 0} B_n \frac{x^n}{n!}, \quad \frac{1}{1-e^x} = -\frac{1}{x} + \frac{1}{2} + \dots$$

Thus,

$$\frac{1}{1-e^{-2\pi y}} = \frac{1}{2\pi y} + h(y)$$

Differentiating m times then

$$(-2\pi)^m \sum_{m \geq 1} n^m e^{-2\pi n y} = \frac{1}{2\pi} \frac{(-1)^m}{y^{m+1}} + h^m(y) \sim y^{-(m+1)}$$

From above computation $|f(iy)| \sim y^{-(A+1)}$ as $y \rightarrow 0$. We can apply previous Theorem with $\sigma > 1 + A$. Take $\sigma > \max(1 + A, k/2)$:

$$\begin{aligned} f(iy) &= \phi(y) = \frac{1}{2\pi i} \int_{\text{Re}(s)=\sigma} \mathcal{M}(s) y^{-s} ds \\ &= \frac{1}{2\pi i} \int_{\text{Re}(s)=k/2} \mathcal{M}(s) y^{-s} ds \\ &= \frac{1}{2\pi i} \int_{\text{Re}(s)=k/2} (-1)^{k/2} \mathcal{M}(k-s) y^{-s} ds \\ &= \frac{1}{2\pi i} (-1)^{k/2} \int_{\text{Re}(s)=k/2} \mathcal{M}(s) y^{-(k-s)} ds \\ &= \frac{1}{2\pi i} \left(\frac{i}{y}\right)^k \int_{\text{Re}(s)=k/2} \mathcal{M}(s) \left(\frac{1}{y}\right)^{-s} ds \\ &= \left(\frac{i}{y}\right)^k f\left(\frac{i}{y}\right) \\ &= (-1)^k / (iy)^k f\left(-\frac{1}{iy}\right) = \frac{1}{(iy)^k} f\left(-\frac{1}{iy}\right) \end{aligned}$$

Since $|\mathcal{M}(s) = \mathcal{M}(k-s)|$, and $|\mathcal{M}(s)| \sim 1/\text{Im}^2(s)$ as $\text{Im}(s) \rightarrow \pm\infty$. Thus, $|\mathcal{M}(s)| \sim 1/\text{Im}^2(s)$ as $\text{Im}(s) \rightarrow \pm\infty$ uniformly on $\{s \in \mathbb{C} \mid k - \sigma \leq \text{Re}(s) \leq 0\}$ which is known as the Phragmen–Lindelof principle. $\square$

8 Applications to Quadratic Forms

Definition 8.1. A quadratic form is a homogeneous polynomial of degree 2 and we will always consider integer coefficients.

Examples. Examples of quadratic forms are

$$Q(x, y) = x^2 + y^2, \quad Q(x, y) = x^2 + 3xy + 2y^2, \quad Q(x_1, x_2, x_3, x_4) = x_1^2 + 5x_2x_3 + 6x_4^2$$

Main Question: In how many integers can $m \in \mathbb{Z}$ be represented by a quadratic form $Q(x_1, \dots, x_n)$. We want to find

$$r_Q(m) = \#\{(x_1, \dots, x_n) \in \mathbb{Z}^n \mid Q(x_1, \dots, x_n) = m\}$$

Idea: look at $\sum_{m \in \mathbb{Z}} r_Q(m) q^m$ and we would like to find an explicit generating function for $r_Q(m)$. We would look at cases when we end up with modular forms for $SL_2(\mathbb{Z})$. First, some backgrounds:

Definition 8.2 (Lattices in $\mathbb{R}^n$). $\Lambda \subseteq \mathbb{R}^n$ is a lattice if $\Lambda = \mathbb{Z}v_1 + \dots + \mathbb{Z}v_n$ where $\{v_1, \dots, v_n\}$ is a basis of $\mathbb{R}^n$.

Example. $\mathbb{Z}^n = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_n$ is a lattice. $t\mathbb{Z}^n$ with $t \in \mathbb{R}_{\neq 0}$ is a lattice.

In general, $\Lambda = \mathcal{B}_\Lambda \mathbb{Z}^n$ where

$$\mathcal{B}_\Lambda = (v_1 \ \dots \ v_n)$$

where $\{v_1, \dots, v_n\}$ is a basis of $\mathbb{R}^n$. Let $A_\Lambda = \mathcal{B}_\Lambda^T \mathcal{B}_\Lambda = (a_{ij})_{1 \leq i, j \leq n}$ where $a_{ij} = v_i^T v_j = v_i \cdot v_j$ so $a_{ij} = a_{ji}$. So A_Λ is a symmetric matrix and $\det(\mathcal{B}_\Lambda) \neq 0$ since entries of columns are basis i.e. $\det(A_\Lambda) > 0$. Now consider

$$\text{covol}(\Lambda) = \text{vol}\{c_1 v_1 + \dots + c_n v_n \mid c_i \in [0, 1], i = 1, \dots, n\} = |\det(\mathcal{B}_\Lambda)|$$

The dual lattice of Λ with respect to the basis $\{v_1, \dots, v_n\}$ is

$$\Lambda^\vee := \{x \in \mathbb{R}^n \mid x \cdot v \in \mathbb{Z}, \forall v \in \Lambda\}$$

Example. $(\mathbb{Z}^n)^\vee = \mathbb{Z}^n$ and $(t\mathbb{Z}^n)^\vee = \frac{1}{t}\mathbb{Z}^n$.

Is $\Lambda^\vee$ actually a basis? For each $i = 1, \dots, n$ let $\omega_i \in \mathbb{R}^n$ be the unique vector in $\mathbb{R}^n$ such that

$$\omega_i \cdot v_j = \delta_{ij}$$

Lemma 8.3. $\Lambda^\vee = \mathbb{Z}\omega_1 + \dots + \mathbb{Z}\omega_n$

Proof. Let $x \in \mathbb{R}^n$ then $x = \sum_{i=1}^n c_i \omega_i$ for $c_i \in \mathbb{R}$ since $\{\omega_1, \dots, \omega_n\}$ is a basis. $v \in \Lambda$ if and only if $v = \sum_{j=1}^n d_j v_j$ with $d_j \in \mathbb{Z}$. $x \in \Lambda^\vee$ if and only if $x \cdot v \in \mathbb{Z}$ for all $v \in \Lambda$ i.e.

$$\begin{aligned} & \left(\sum_{i=1}^n c_i \omega_i \right) \cdot \left(\sum_{j=1}^n a_j v_j \right) \in \mathbb{Z}, \quad \forall d_j \in \mathbb{Z}, j = 1, \dots, n \\ \implies & \left(\sum_{i=1}^n \sum_{j=1}^n c_i a_j \omega_i \cdot v_j \right) \in \mathbb{Z}, \quad \forall d_j \in \mathbb{Z}, j = 1, \dots, n \\ & \sum_{i=1}^n c_i d_i \in \mathbb{Z}, \quad \forall d_i \in \mathbb{Z} \end{aligned}$$

Therefore, we conclude that $c_i \in \mathbb{Z}$ for all $i = 1, \dots, n$. Thus, $x \in \mathbb{Z}\omega_1 + \dots + \mathbb{Z}\omega_n$. □

Remark. Let $\mathcal{B}_{\Lambda^\vee} := (\omega_1, \dots, \omega_n)$. Then $(\mathcal{B}_{\Lambda^\vee})^T \mathcal{B}_\Lambda = I_{n \times n} \implies \mathcal{B}_{\Lambda^\vee} = (\mathcal{B}_\Lambda^{-1})^T$. And $A_{\Lambda^\vee} = (\mathcal{B}_{\Lambda^\vee})^T \mathcal{B}_{\Lambda^\vee} = \mathcal{B}_\Lambda^{-1} (\mathcal{B}_\Lambda^{-1})^T = (\mathcal{B}_\Lambda^T \mathcal{B}_\Lambda)^{-1} = A_\Lambda^{-1}$.

Lemma 8.4. Λ is self-dual ($\Lambda = \Lambda^\vee$) if and only if $A_\Lambda \in SL_n(\mathbb{Z})$.

Proof. Suppose $\Lambda = \Lambda^\vee$ then $\Lambda \subset \Lambda^\vee$ implies

$$\{x \cdot v \in \mathbb{Z}, \quad \forall x, v \in \Lambda\}$$

In particular $v_i \cdot v_j \in \mathbb{Z}$ for all $i, j = 1, \dots, n$ which means $A_\Lambda \in M_n(\mathbb{Z})$. We also have $\Lambda^\vee \subset \Lambda = (\Lambda^\vee)^\vee$ therefore we see that $A_{\Lambda^\vee} \in M_n(\mathbb{Z})$. But we see previously that $A_{\Lambda^\vee} = A_\Lambda^{-1} \in M_n(\mathbb{Z})$. Therefore, $\det(A_\Lambda) \det(A_\Lambda^{-1}) = 1$ and determinant must both be a positive integer thus $A_\Lambda \in SL_n(\mathbb{Z})$.

Conversely, suppose $A_\Lambda \in SL_n(\mathbb{Z})$ then $v_i \cdot v_j \in \mathbb{Z}$ for all $i, j = 1, \dots, n$ which implies $v_i \in \Lambda^\vee$ for all i then $\Lambda = \mathbb{Z}v_1 + \dots + \mathbb{Z}v_n \subseteq \Lambda^\vee$. In addition we see that $A_\Lambda \in SL_n(\mathbb{Z}) \implies A_\Lambda^{-1} = A_{\Lambda^\vee} \in SL_n(\mathbb{Z})$. Similar to above, we conclude that $\Lambda^\vee \subseteq (\Lambda^\vee)^\vee = \Lambda$. □

Remark. Note if $\Lambda = B_\Lambda \mathbb{Z}^n$ then $\Lambda^\vee = (B_\Lambda^{-1})^T \mathbb{Z}^n$.

A self-dual lattice $\Lambda \subseteq \mathbb{R}^n$ gives rise an integral quadratic form:

$$\begin{aligned} Q_\Lambda(x_1, \dots, x_n) &= \|x_1 v_1 + \dots + x_n v_n\|^2 \\ &= (x_1 v_1 + \dots + x_n v_n)^T (x_1 v_1 + \dots + x_n v_n) \\ &= (x_1, \dots, x_n) B_\Lambda^T B_\Lambda (x_1, \dots, x_n)^T \\ &= (x_1, \dots, x_n) A_\Lambda (x_1, \dots, x_n)^T \end{aligned}$$

and $A^\vee \in SL_n(\mathbb{Z})$ since $\Lambda = \Lambda^\vee$. Then $r_{Q_\Lambda}(m) = \#$ of vectors in Λ of length squared $= m$. So $r_{Q_\Lambda}(m) = 0$ if $m < 0$. $r_{Q_\Lambda}(m) = 1$ and $r_{Q_\Lambda}(m)$ is finite, $r_{Q_\Lambda}(m) = O(m^{1/2})$. We associate a theta series to Λ/Q_Λ

$$\Theta_\Lambda(z) := \sum_{x \in \Lambda} e^{\pi i \|x\|^2 z} = \sum_{x \in \Lambda} q^{\|x\|^2/2}, \quad z \in \mathbb{H}, q = e^{2\pi i z}$$

We will make a further assumption of Λ ; it's even i.e. $x \cdot x \in 2\mathbb{Z}$ for all $x \in \Lambda$. Then

$$\Theta_\Lambda(z) = \sum_{m \geq 0} r_{Q_\Lambda}(2m) q^m$$

Theorem 8.5. If $\Lambda \subseteq \mathbb{R}^n$ is a self-dual, even lattice then $8|n$ and $\Theta_\Lambda(z)$ is a modular form of weight $\frac{n}{2}$.

Proof. We will show Θ_Λ is a modular form and $8|n$ is a by product of the proof of this fact. Θ_Λ is holomorphic on $\mathbb{H}$ by the q -expansion since the coefficients have polynomial growth. $\Theta(z+1) = \Theta(z)$ for all $z \in \mathbb{H}$ by the q expansion. Left to show $\Theta(-\frac{1}{z}) = z^{n/2} \Theta_\Lambda(z)$. Enough to show above for $z = iy, y \in \mathbb{R}_{>0}$ i.e.

$$\begin{aligned} \Theta_\Lambda(-\frac{1}{iy}) &= (iy)^{n/2} \Theta_\Lambda(iy) \\ \sum_{x \in \Lambda} e^{-\pi/y \|x\|^2} &= (iy)^{n/2} \sum_{x \in \Lambda} e^{\pi i \|x\|^2 iy} = (iy)^{n/2} \sum_{x \in \Lambda} e^{-\pi y \|x\|^2} \end{aligned}$$

This follows from Poisson Summation in $\mathbb{R}^n$. If $f \in L^1(\mathbb{R}^n)$ then

$$\hat{f}(x) = \int_{\mathbb{R}^n} f(t) e^{-2\pi i x \cdot t} dt$$

If $f, \hat{f} \in L^1(\mathbb{R}^n)$, f is continuous on $\mathbb{R}^n$ then

$$\sum_{x \in \Lambda} f(x) = \frac{1}{\text{covol}(\Lambda)} \sum_{x \in \Lambda^\vee} \hat{f}(x)$$

as long as both sides are absolutely and uniformly convergent. But since Λ is self-dual so

$$\sum_{x \in \Lambda} f(x) = \sum_{x \in \Lambda} \hat{f}(x)$$

Let $f(x) = e^{-\pi x \cdot x}$ and $g(x) = f(\sqrt{y}x)$ then

$$\Theta_\Lambda(iy) = \sum_{x \in \Lambda} g(x)$$

then $f(x) = \hat{f}(x)$. And $\hat{g}(x) = (\frac{1}{\sqrt{y}} \hat{f})^n(\frac{x}{\sqrt{y}})$ thus

$$\sum_{x \in \Lambda} \hat{g}(x) = (\frac{1}{\sqrt{y}})^n \Theta_\Lambda(-\frac{1}{iy}) = \Theta_\Lambda(iy)$$

We have

$$\Theta_\Lambda(-\frac{1}{z}) = (\frac{z}{i})^{n/2} \Theta_\Lambda(z)$$

Since Θ_Λ is translation invariant then

$$\Theta(1 - \frac{1}{z}) = \Theta(-\frac{1}{z}) = (\frac{z}{i})^{n/2} \Theta(z)$$

Note that $(TS)^3 z = z$ we get

$$\begin{aligned} (\frac{(TS)^2 z}{i})^{n/2} (\frac{(TS)z}{i})^{n/2} (\frac{z}{i})^{n/2} &= 1 \\ \implies [-\frac{1}{i^3}]^{n/2} &= 1 \end{aligned}$$

so $8|n$. And in this case we have

$$\Theta(-\frac{1}{z}) = z^{n/2} \Theta_\Lambda(z)$$

□

Corollary 8.6. If $\Lambda \subset \mathbb{R}^n$ is a self-dual, even lattice, then

$$\Theta_\Lambda = E_{n/2} + f,$$

where $f \in S_{n/2}(SL_2(\mathbb{Z}))$ and $E_{n/2}$ is the normalized Eisenstein series of weight $n/2$. Furthermore, for $m \in \mathbb{N}$,

$$r_\Lambda(2m) = -\frac{n}{B_{n/2}} \sigma_{n/2-1}(m) + O(m^{n/4}).$$

Proof. Recall that

$$E_{n/2} = 1 - \frac{n}{B_{n/2}} \sum_{m \geq 1} \sigma_{n/2-1}(m) q^m.$$

As $r_{Q_\Lambda}(0) = 1$ and $\Theta_\Lambda \in M_{n/2}(SL_2(\mathbb{Z}))$, we must have

$$\Theta_\Lambda - E_{n/2} \in S_{n/2}(SL_2(\mathbb{Z})).$$

The second part now follows from the estimate for the m^{th} Fourier coefficient of a cusp form of weight $n/2$. $\square$

For $n = 8$, it is known that there is only one self-dual, even lattice in $\mathbb{R}^8$ up to isomorphism: Λ_8 . Then

$$\Theta_{\Lambda_8} = E_4 = 1 + 240 \sum_{m \geq 1} \sigma_3(m) q^m.$$

So

$$r_{Q_{\Lambda_8}}(2m) = 240\sigma_3(m), \quad \forall m \in \mathbb{N}.$$

For $n = 16$, we have two non-equivalent self-dual even lattices: $\Lambda_8 \oplus \Lambda_8$ and Λ_{16} . Their corresponding theta functions coincide and equal

$$E_8 = 1 + 480 \sum_{m \geq 1} \sigma_7(m) q^m.$$

since there are no cusp forms of weight 8. Hence, we have

$$r_{Q_{\Lambda_8 \oplus \Lambda_8}}(2m) = r_{Q_{\Lambda_{16}}}(2m) = 480\sigma_7(m), \quad \forall m \in \mathbb{N}.$$

The theta function for $\Lambda_8 \oplus \Lambda_8$ is equal to the square of theta function for Λ_8 and so we recover the identity $E_4^2 = E_8$.

For $z \in \mathbb{H}$, let $\Theta(z) = \sum_{m \in \mathbb{Z}} e^{\pi i m^2 z} = 1 + 2 \sum_{m \geq 1} q^{m^2/2}$ then $\Theta(2z) := \sum_{m \in \mathbb{Z}} q^{m^2}$ and

$$\Theta(2z)^4 = \sum_{x_1, x_2, x_3, x_4 \in \mathbb{Z}} q^{x_1^2 + x_2^2 + x_3^2 + x_4^2} = \sum_{m \geq 0} r_4(m) q^m$$

Now notice that $\Theta(iy) = \sum_{m \in \mathbb{Z}} e^{-\pi m^2 y} = \Theta(y)$. We know $\Theta(\frac{1}{y}) = \sqrt{y} \Theta(y)$. So $\Theta(i/y) = \sqrt{y} \Theta(iy)$, for all $y \in \mathbb{R}_{>0}$ and $\Theta(-1/iy) = \sqrt{y} \Theta(iy)$ then we get

$$\begin{aligned} \Theta\left(-\frac{1}{z}\right) &= \sqrt{\frac{z}{i}} \Theta(z), \quad \forall z \in \mathbb{H} \\ \implies \Theta\left(-\frac{1}{z}\right)^4 &= -z^2 \Theta(z)^4, \quad \forall z \in \mathbb{H} \end{aligned}$$

Theorem 8.7. $f := \Theta(2z)^4 \in M_2(\Gamma_0(4))$.

Proof. We need to know more about $\Gamma_0(4)$:

(1) $\Gamma_0(4)$ is generated by $\pm I$, T and $U = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$

(2) It has 3 cusps: $\infty, 0$ and $\frac{1}{2}$.

We have to check the modularity condition with weight 2 for U note that

$$U = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}^{-1}$$

Denote the first matrix by W then

$$f|_2 U = (f|_2 W_4)|_2 T^{-1}|_2 \left(-\frac{1}{4W_4}\right) = (f|_2 W_4)|_2 T^{-1}|_2 W_4$$

$$f|_2 W_4 = 4(4z)^{-2} f(-1/4z) = \frac{1}{4z^2} \Theta(2(-\frac{1}{4z}))^4 = \frac{1}{4z^2} \Theta(-\frac{1}{2z})^4 = \frac{1}{4z^2} (-(2z)^2) \Theta(2z)^4 = -f$$

which shows $f|_2 U = f$. f is holomorphic at $i\infty$ by q -expansion. At 0, notice $W_4(i\infty) = 0$. At $1/2$, note that

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} i\infty = \frac{1}{2} \text{ and}$$

$$V = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = W_2 T^{-1} W_2^{-1} \text{ where } W_2 = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$$

We need to show $f|_V$ is holomorphic at $i\infty$. $f|_2 W_2 = 2(2z)^{-2} f(-1/2z) = \frac{1}{2z^2} \Theta(-1/2z)^4 = \frac{1}{2z^2} (-z^2) \Theta(z)^4 = -\frac{1}{2} \Theta(z)^4$.

Let

$$\Theta_3(z) := \Theta(z-1) = \sum_{m \in \mathbb{Z}} e^{\pi i m^2 (z-1)} = \sum_{m \in \mathbb{Z}} (-1)^m e^{\pi i m^2 z}$$

Note that $\Theta_3(z)^4 = -z^2 \Theta_2(z)^4$ where $\Theta_2(z) = \sum_{m \in \mathbb{Z}} e^{\pi i (m+1/2)^2 z}$. If you replace m by $-m-1$ then you get the same thing. Thus,

$$\Theta_2(z) = 2q^{1/8} \sum_{m \geq 0} q^{m(m+1)/2}$$

$(f|_2 W_2)|_2 T^{-1} = (-\frac{1}{2} \Theta(z)^4)|_2 T^{-1} = -\frac{1}{2} \Theta_3(z)^3$ and

$$(f|_2 V)|_2 W_2 = (-\frac{1}{2} \Theta_3(z)^3)|_2 W_2 = (-\frac{1}{2}) 2(2z)^{-2} \Theta_3(-\frac{1}{2z})^4 = -\frac{1}{(2z)^2} (-(2z)^2) \Theta_2(2z)^4 = \Theta_2(2z)^4$$

which is holomorphic at $i\infty$. Fact: $\dim_{\mathbb{C}}(\Gamma_0(4)) \leq 2$. $\square$

Proposition 8.8. Let $g(z) = 2E_2(2z) - E_2(z)$ where E_2 is the Eisenstein series of weight 2. And $h(z) = \frac{1}{2}g|_2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = g(2z)$. Then $g, h \in M_2(\Gamma_0(4))$ and they are $\mathbb{C}$ -linearly independent.

Proof. We will show that $g \in M_2(\Gamma_0(2)) \subset M_2(\Gamma_0(4))$ and hence $h = \frac{1}{2}g|_2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \in M_2(\Gamma_0(4))$. $\Gamma_0(2)$ is generated by $\pm I$, T and $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ and notice $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = W_2 T^{-1} W_2^{-1}$ where $\begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$. g is T -invariant since E_2 is. Recall $E_2(-1/2) = z^2 E_2(z) - \frac{6i}{\pi} z$. Now,

$$\begin{aligned} (g|_2 W_2)(z) &= 2(2z)^{-2} g(-\frac{1}{2z}) = \frac{1}{2z^2} [2E_2(-\frac{1}{z}) - E_2(-\frac{1}{2z})] \\ &= \frac{1}{2z^2} [2(z^2 E_2(z) - \frac{6iz}{\pi}) - ((2z)^2 E_2(2z) - \frac{6i}{\pi} \cdot 2z)] \\ &= E_2(z) - 2E_2(2z) = -g \end{aligned}$$

Thus,

$$\begin{aligned} g|_2(W_2 T^{-1} W_2^{-1}) &= (-g|_2 T^{-1})|_2 W_2 \\ &= (-g)|_2 W_2 = -(-g) = g \end{aligned}$$

We need to check the behaviour at the cusps of $\Gamma_0(2) : 0, \infty$. g is holomorphic at ∞ since E_2 is. At 0, notice that $0 = W_2 \infty$ so g is holomorphic at 0 if and only if $g|_2 W_2$ is holomorphic at ∞ but this is $-g$ so no problem here.

$g(z) = 2E_2(2z) - E_2(z)$ where $E_2 = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$. Thus

$$\begin{aligned} g(z) &= 2(1 - 24 \sum_{n \geq 1} \sigma_1(n) q^{2n}) - (1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n) \\ &= 1 + \sum_{m \geq 1} q^n (24\sigma_1(n) - 2 \cdot 24\sigma_1(\frac{n}{2})) \\ &= 1 + 24q + 24q^2 + \dots \end{aligned}$$

Additionally, $h = g(2z) = 1 + 24q^2 + 24q^4$ where we see that g and h are not $\mathbb{C}$ -proportional. In particular, $f = ag + bh$ for some $a, b \in \mathbb{C}$ where $f = \Theta(2z)^4 = 1 + (1 + 2 \sum_{n \geq 1} q^{n^2})^4 = 1 + 8q + \dots$. Compare coefficient we have $a = 1/3$ and $b = 2/3$.

$$\begin{aligned} f &= \frac{1}{3}g + \frac{2}{3}h \\ &= \frac{1}{3}(1 + 24 \sum_{n \geq 1} q^n \sigma_1(n) - 2\sigma_1(\frac{n}{2})) + \frac{2}{3}(1 + 24 \sum_{n \geq 1} q^{2n} (\sigma_1(n) - 2\sigma_1(\frac{n}{4}))) \\ &= 1 + 8 \sum_{n \geq 1} q^n (\sigma_1(n) - 4\sigma_1(\frac{n}{4})) \\ &= 1 + 8 \sum_{\substack{d|n \\ d>0}} d - 4 \sum_{\substack{d|n/4 \\ d>0}} d \\ &= 1 + 8 \sum_{n \geq 1} q^n (\sum_{\substack{d|n \\ 4 \nmid d}} d) \end{aligned}$$

Thus, $r_4(m) = 8 \sum_{n \geq 1} q^n (\sum_{\substack{d|n \\ 4 \nmid d}} d)$ for $m \in \mathbb{Z}$. □

Remark. We claimed $\Gamma_0(4)$ is generated by $\pm I, T$ and $U = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(4)$. Since $\det(\gamma) = 1$ so a is odd. Look at $a^2 + b^2$ and $a^2 + b^2 \geq 1$ and we have equality if and only if $a = \pm 1$ and $b = 0$ where $d = \pm 1$ i.e. $\gamma = \pm U^n$.

$$\gamma T^{\pm 1} = \begin{pmatrix} a & b \\ 4c & d \end{pmatrix} \begin{pmatrix} 1 & \pm 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & \pm a + b \\ 4c & \pm 4c + d \end{pmatrix}$$

and

$$\gamma U^{\pm 1} = \begin{pmatrix} a & b \\ 4c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \pm 4 & 1 \end{pmatrix} = \begin{pmatrix} a \pm 4b & b \\ 4c \pm 4d & d \end{pmatrix}$$

Compare $a^2 + (\pm a + b)^2$ and $(a \pm 4b)^2 + b^2$ with $a^2 + b^2$.

$$a^2 + (\pm a + b)^2 \leq a^2 + b^2 \Leftrightarrow a^2 \leq \pm 2ab$$

and

$$(a \pm 4b)^2 + b^2 \leq a^2 + b^2 \Leftrightarrow 2b^2 \leq \pm ab$$

If $|a| < 2|b|$ then apply $T^{\pm 1}$. If $|a| \geq 2|b|$ apply $U^{\pm 1}$.

Theorem 8.9. Let $\Gamma \subseteq SL_2(\mathbb{Z})$ be a congruence subgroup, let $k \in \mathbb{Z}$. Then $\dim_{\mathbb{C}} M_k(\Gamma) \leq 1 + \lfloor \frac{Dk}{12} \rfloor$ where $D = [SL_2(\mathbb{Z}) : \Gamma]$, $k \geq 0$ and $\dim_{\mathbb{C}} M_k(\Gamma) = 0$ if $k < 0$.

Proof. Let $SL_2(\mathbb{Z}) = \bigcup_{i=1}^D \Gamma \gamma_i$. Let $f \in M_k(\Gamma)$ and let $N_f = \prod_{i=1}^D f|_k \gamma_i$. N_f is well-defined, i.e. it does not depend on the choice of representatives γ_i . Since

$$f|_k(\gamma \gamma_i) = (f|_k \gamma)|_k \gamma_i = f|_k \gamma_i$$

We claim that $N_f \in M_{kD}(SL_2(\mathbb{Z}))$.

Proof for the Claim. Let $\gamma \in SL_2(\mathbb{Z})$, $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\begin{aligned} (N_f|_{kD} \gamma)(z) &= (cz + d)^{-kD} N_f(\gamma z) \\ &= (cz + d)^{-kD} \prod_{i=1}^D (f|_k \gamma_i)(\gamma z) \\ &= \prod_{i=1}^D (cz + d)^{-k} (f|_k \gamma_i)(\gamma z) \\ &= \prod_{i=1}^D (f|_k \gamma_i)|_k \gamma \\ &= \prod_{i=1}^D (f|_k \gamma_i \gamma)(z) = N_f(z) \end{aligned}$$

Multiplication by $\gamma \in SL_2(\mathbb{Z})$ on the right permutes the right coset of Γ in $SL_2(\mathbb{Z})$: For each i , there exists unique γ such that $\gamma_i \gamma = \gamma_0^j \gamma_j$ for some $\gamma_0^j \in \Gamma$. Thus,

$$\prod_{i=1}^D (f|_k \gamma_i \gamma) = \prod_{j=1}^D f|_k \gamma_0^j \gamma_j = \prod_{j=1}^D (f|_k \gamma_j) = N_f$$

N_f is a product of holomorphic function so it must be holomorphic as well. $\square$

Now notice $N_f = 0$ if and only if $f = 0$. Since $M_k(\Gamma) = \{0\}$ for $k < 0$. Suppose $N_f \neq 0$, by the Valence Formula for all $p \in \mathbb{H}/SL_2(\mathbb{Z})$, $v_p(N_f) \leq \frac{kD}{12}$. Since $v_p(N_f) = \sum_{i=1}^D v_p(f|_k \gamma_i)$ we have $w_p(f) \leq \frac{kD}{12}$ for all $p \in \mathbb{H}/SL_2(\mathbb{Z})$ and p not equivalent to i and ρ . Let $M = 1 + \lfloor \frac{kD}{12} \rfloor$, let $P_1, \dots, P_M$ be distinct points in the $\mathbb{H}/SL_2(\mathbb{Z}) \setminus \{i, \rho\}$. Let $f_1, \dots, f_{M+1} \in M_k(\Gamma)$ then there exists a linear combination of the f_i 's,

$$f = \sum_{i=1}^{M+1} c_i f_i, \quad c_i \in \mathbb{C}$$

such that $f(P_j) = 0$ for all $j = 1, \dots, M$. This is by linear algebra, plug all the P_j 's into f we get a system of $M+1$ linear homogeneous equations with M unknowns. So $f \in M_k(\Gamma)$ and has at least M zeros $> \frac{kD}{12}$. From the Valence Formula, applied to N_f , we also have

$$\sum_{p \in \mathbb{H}/SL_2(\mathbb{Z}) \setminus \{i, \rho\}} v_p(f) \leq \frac{kD}{12}$$

i.e. $f = 0$. Thus, in conclusion $f_1, \dots, f_{M+1}$ must be linearly independent. Since they are arbitrary, any $M+1$ modular forms in $M_k(\Gamma)$ are linearly dependent thus,

$$\dim_{\mathbb{C}}(M_k(\Gamma)) \leq 1 + \lfloor \frac{Dk}{12} \rfloor$$

$\square$

Example. $M_2(\Gamma_0(4))$, we know that $[SL_2(\mathbb{Z}) : \Gamma_0(4)] = 6$ thus,

$$\dim_{\mathbb{C}}(M_2(\Gamma_0(4))) \leq 1 + \lfloor \frac{6 \times 2}{12} \rfloor = 2$$